



CHAPTER 1 THE PROBLEM

Introduction

The increasing reliance on electrical appliances in residential settings has led to higher energy consumption and rising electricity costs worldwide. In the Philippines, residential electricity rates remain among the highest in Southeast Asia, making energy monitoring an important concern for households, tenants, and landlords. The average Filipino household consumes approximately 200–400 kilowatt-hours (kWh) per month, and electricity expenses form a significant portion of household budgets.

However, conventional power meters only provide total energy consumption and do not present detailed information regarding individual appliance usage. This limitation makes it difficult for residents—particularly tenants in boarding houses and rental units—to determine which appliances consume the most electricity and how usage can be optimized. As a result, energy awareness remains limited, and fair cost allocation becomes challenging.

Non-Intrusive Load Monitoring (NILM) offers a solution by analyzing electrical consumption patterns to identify appliance usage from aggregated energy data without requiring sensors on each device. NILM has evolved from early research into practical software-based approaches capable of classifying appliances using data analytics, pattern recognition, and machine learning techniques.

With the advancement of backend and mobile technologies, software-based NILM platforms can provide near real-time monitoring, remote access, and intelligent visualization. Centralized databases such as MySQL support scalable storage and efficient retrieval of consumption records, while mobile applications allow users to monitor usage anytime and anywhere.

This study focuses on the design and development of a software-based NILM system for residential energy monitoring. The system processes electrical parameter data obtained from existing smart meters, third-party IoT sources, or simulated datasets. The monitoring hardware and electrical data acquisition components were developed in collaboration with the Electrical Engineering team, while this study primarily concentrates on software processing, appliance classification, backend integration, and mobile application development. Through backend server processing and a cross-platform mobile application, the system enables tenants to monitor energy consumption and identify high-usage appliances, while landlords can oversee unit-level electricity usage to support fair billing and improved energy awareness.

Statement of the Problem

This study addresses the limitations of conventional residential energy monitoring by developing a software-based NILM system capable of providing appliance-level identification and real-time energy consumption insights for residential settings, particularly boarding houses and rental properties.

General Problem

How can a software-based Non-Intrusive Load Monitoring (NILM) system be designed, developed, and evaluated to provide appliance classification and real-time residential energy monitoring?

Specific Problems

1. How can the system classify household appliances using aggregated energy consumption data without requiring intrusive per-appliance sensors?
2. How can the system enhance tenant energy awareness through real-time visualization of instantaneous power and accumulated energy usage?
3. How can the system provide landlords with a centralized platform for monitoring unit-level electricity consumption and supporting fair billing?

Conceptual Framework (IPO Model)

The study adopts the Input–Process–Output (IPO) Model as the conceptual framework in designing and developing the Software-Based Non-Intrusive Load Monitoring (NILM) System. The model explains how energy consumption datasets are collected, processed, and transformed into meaningful information for residential energy monitoring and appliance classification.

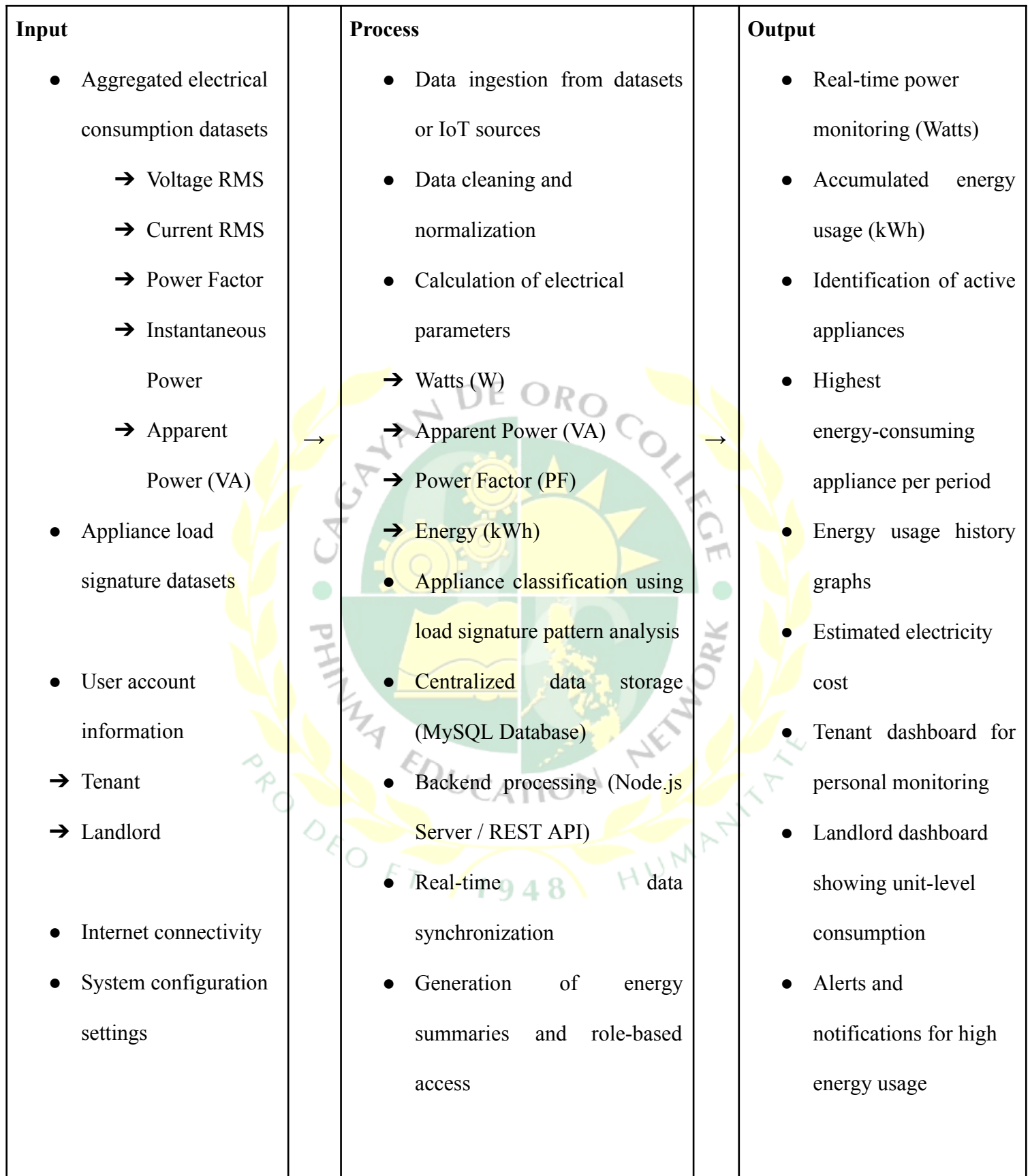


Figure 1. Input–Process–Output Model of the Proposed Software-Based NILM System



- **Input** The input stage includes the data and components required for system operation, such as aggregated electrical consumption datasets (Voltage RMS, Current RMS, Power Factor, Instantaneous Power, and Apparent Power), appliance load signature datasets, user account information (Tenant and Landlord roles), Internet connectivity, and system configuration settings.
- **Process** The process stage involves how the system transforms input data into usable information, including data ingestion from available datasets, data filtering and normalization, calculation of electrical parameters (W, VA, PF, and kWh), appliance classification using load signature pattern analysis, data storage in the centralized MySQL database, backend processing using Node.js (REST API) services, real-time data synchronization, generation of energy consumption summaries, and role-based data access for tenants and landlords.
- **Output** The output stage presents the information produced by the system, such as real-time power monitoring (Watts), accumulated energy usage (kWh), identification of active appliances, highest energy-consuming appliance per period, energy usage history graphs, estimated electricity cost, landlord dashboard showing unit-level consumption, and alerts and notifications for high energy usage.



Objectives of the Study

General Objective

To design, develop, and evaluate a software-based NILM system for residential energy monitoring.

Specific Objectives

1. To develop an appliance classification module using aggregated electrical consumption data (NILM load signatures).
2. To design and develop a mobile application for real-time monitoring, visualization, and user dashboards.
3. To evaluate the software system in terms of usability, accuracy, reliability, and performance efficiency.

Significance of the Study

Tenants – Provides awareness of personal energy consumption and helps identify high-usage appliances.

Landlords – Supports transparent monitoring and fair electricity cost allocation.

Researchers – Serves as reference for software-based NILM and IoT energy analytics.

Academic Community – Demonstrates integration of mobile development, cloud computing, and data analytics.



Scope and Limitations of the Study

This study focuses on the design, development, and evaluation of a software-based Non-Intrusive Load Monitoring (NILM) system for residential environments, particularly boarding houses and rental units. The system processes aggregated electrical consumption datasets containing parameters such as Voltage RMS, Current RMS, Instantaneous Power (W), Apparent Power (VA), Power Factor (PF), and estimated energy consumption (kWh) generated from external monitoring hardware.

The collected data are stored in a centralized MySQL database and processed through backend services developed using Node.js. A mobile application developed using React Native and Expo provides real-time monitoring, historical consumption visualization, estimated electricity cost computation, role-based dashboards, and alert notifications for tenants and landlords. The system detects appliance usage patterns and identifies unusual or prolonged electricity consumption.

The study also includes remote outlet control functionality that allows users to switch power outlets ON or OFF through the mobile application. Control commands are transmitted to the NILM monitoring device, where hardware switching components execute the outlet switching operation. Appliances connected to the controlled outlet are therefore indirectly managed. The mobile application serves as the primary interface for both monitoring and control.

However, the study is limited to residential energy consumption datasets and does not cover industrial or large-scale monitoring scenarios. Appliance identification is restricted to selected common household appliances and may not generalize across all appliance brands or models. Classification is limited to ON/OFF detection and signature-based analysis using



available datasets, and computed energy values represent estimates rather than utility-grade measurements.

System accuracy may be affected by appliances with similar consumption patterns and simultaneous appliance operation. Real-time monitoring and remote control performance depend on Internet connectivity, server availability, and the operational status of the monitoring device and switching components. The electrical monitoring hardware, sensor calibration, and switching circuitry were developed in collaboration with the Electrical Engineering team and are not the primary focus of this study. Instead, the research emphasizes the software layer, including NILM processing, backend services, data analytics, and mobile application development. Finally, system evaluation is limited to the defined testing period and selected respondents.

The system may have difficulty accurately identifying or classifying individual appliances when multiple devices are connected through extension outlets or power strips. Since several appliances share a single electrical pathway, their load signatures can overlap, which reduces classification accuracy.

Definition of Terms

Appliance Classification – The process of identifying household appliances by analyzing aggregated electrical consumption patterns using the NILM system.

Apparent Power (VA) – The total electrical power combining real and reactive components used in NILM analysis.

Cloud Computing – The use of Internet-hosted services for data storage, processing, and synchronization, enabling remote access to the system.



Backend Services (Node.js) – Server-side services responsible for processing energy data, validating readings, implementing business logic, and generating system notifications.

Current RMS – The effective current value used for estimating energy consumption.

Dashboard – The user interface that displays real-time energy data, consumption summaries, alerts, and visualization graphs.

Data Visualization – The graphical presentation of energy consumption data through charts, graphs, and dashboards.

Energy Consumption (kWh) – The total amount of electrical energy used over a period of time, measured in kilowatt-hours.

MySQL Database – A relational database used to store, manage, and retrieve energy monitoring data, user information, and system logs.

Instantaneous Power (W) – The amount of electrical power consumed at a specific moment, measured in watts.

ISO/IEC 25010 – An international software quality standard used to assess system performance, usability, reliability, and other quality characteristics.

Landlord/Admin – A user role with access to summarized unit-level energy consumption reports and monitoring features.

Load Signature – The unique energy consumption pattern of an appliance used by the NILM system for identification.

Mobile Application – A cross-platform software application that provides energy monitoring, visualization, and system management features.



Non-Intrusive Load Monitoring (NILM) – A software-based method of identifying appliance usage from aggregated energy consumption data without installing sensors on individual devices.

Power Factor (PF) – A value that indicates how efficiently electrical power is being used, derived from electrical parameter data.

React Native – A software framework used to develop cross-platform mobile applications using a single codebase.

Real-Time Monitoring – The continuous display of updated energy consumption data with minimal delay.

Software-Based NILM System – A platform that processes electrical datasets, performs appliance classification, and provides visualization without requiring dedicated hardware sensors.

System Usability Scale (SUS) – A standardized tool used to evaluate the usability of the developed software system.

Tenant – A user role that allows viewing of personal or unit-level energy consumption and appliance usage insights.

Voltage RMS – The effective voltage value used by the system for power calculations.



CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the review of related literature and studies that establish the theoretical and empirical foundation for the development of the proposed software-based Non-Intrusive Load Monitoring (NILM) system. The review includes both local and foreign literature focusing on energy monitoring platforms, electrical load signature analysis, appliance classification techniques, data analytics, cloud-based architectures, and mobile application integration for real-time monitoring.

The chapter examines the evolution of NILM technology from its foundational principles to modern software-driven approaches that utilize signal processing, pattern recognition, and machine learning. It also discusses how cloud computing services and mobile-first architectures enhance residential energy transparency and accessibility through real-time visualization and analytics.

Furthermore, the review contextualizes the study within the Philippine residential setting, particularly in boarding houses and rental properties where energy cost allocation and consumption awareness remain significant concerns. The chapter concludes with a synthesis of reviewed literature and the identification of research gaps that justify the development of a cost-effective, mobile-integrated NILM software platform tailored to the Philippine context.



Local Literature

Energy Monitoring in the Philippines

The Philippines faces significant challenges in energy consumption management, with residential electricity rates among the highest in Southeast Asia. According to the Department of Energy (DOE), residential electricity consumption has steadily increased, with average household usage ranging from 200–400 kWh per month. The lack of detailed consumption insights limits consumers' ability to identify energy-saving opportunities and adjust usage behavior effectively. Government agencies such as the Energy Regulatory Commission have promoted smart metering initiatives to improve energy efficiency; however, most implementations focus on aggregated consumption data rather than appliance-level insights. Consequently, the application of software-based Non-Intrusive Load Monitoring (NILM) analytics in residential environments remains limited in the Philippine context. Additionally, the Data Privacy Act of 2012 (RA 10173) requires systems handling energy consumption data to ensure proper user consent, secure storage, and controlled access. These requirements are particularly relevant for multi-tenant monitoring platforms where role-based access and data ownership must be carefully managed.

Non-Intrusive Load Monitoring (NILM) was first introduced by Hart (1992), demonstrating that individual appliance consumption could be disaggregated from aggregated household energy data without installing sensors on each device. Over time, NILM has evolved from rule-based analysis toward software-driven approaches that utilize data analytics, pattern recognition, and machine learning to improve classification accuracy (Zoha et al., 2012; Kelly & Knottenbelt, 2015). The fundamental concept of NILM is that each appliance produces a unique load signature characterized by steady-state power behavior, transient events during switching, and time-series consumption patterns. By extracting



relevant features from these patterns, software algorithms can identify appliance operation from overall energy usage. Contemporary NILM approaches include rule-based classification methods, probabilistic models such as Hidden Markov Models, neural network-based learning, deep learning frameworks, and hybrid analytical techniques that combine multiple methods to enhance performance and reliability (Zoha et al., 2012; Kelly & Knottenbelt, 2015).

Recent research emphasizes dataset-based NILM implementations in which appliance identification is performed using historical consumption data rather than dedicated hardware instrumentation (Makonin et al., 2017). Through proper data preprocessing, normalization, and feature extraction, classification models can achieve improved accuracy while remaining computationally efficient (Bonfigli et al., 2018). This shift toward software-centric NILM aligns with the growing availability of smart meter datasets and cloud computing platforms that support large-scale time-series analysis (Makonin et al., 2017).

Cloud-integrated energy monitoring systems play a critical role in enabling scalable NILM deployment. Cloud computing provides flexible data storage, cross-platform accessibility, and cost-effective infrastructure for processing energy data (Hashem et al., 2015). Relational databases such as MySQL support structured storage, efficient querying, and scalable backend integration for software-based NILM platforms, allowing continuous data management and reliable retrieval of monitoring records without requiring complex infrastructure.

Mobile application integration further enhances the usability of NILM systems by serving as the primary interface between users and energy analytics. Mobile platforms provide real-time visualization of consumption data, historical usage analysis, appliance activity insights, and alert mechanisms that notify users of unusual consumption patterns.



Studies indicate that mobile-first monitoring significantly improves user engagement, promotes energy awareness, and encourages behavioral changes that contribute to energy savings (Darby, 2006; Fischer, 2008; Karakose et al., 2020). As a result, the combination of NILM analytics, backend data management, and mobile application development represents a comprehensive software approach to residential energy monitoring.

Local Studies

Energy Monitoring in Philippine Residential Settings

Several studies conducted in the Philippine context highlight the importance of detailed energy monitoring and the growing demand for non-intrusive, mobile-based solutions. Dela Cruz and Santos (2020) analyzed energy consumption patterns among 500 households in Metro Manila and found that air conditioning units account for 40–50% of total consumption, refrigerators consume 15–20%, while lighting and other appliances contribute approximately 30–35%, with peak usage occurring during evening hours and average monthly consumption ranging from 200–400 kWh. Their findings emphasized the need for appliance-level monitoring to help consumers identify energy-saving opportunities. Garcia and Tan (2021) examined smart home technology adoption barriers and reported limited awareness of energy monitoring technologies, cost concerns as a major barrier, strong preference for mobile-based monitoring solutions, and the need for non-intrusive installation methods with privacy protections. Similarly, Villanueva and Ramos (2019) explored energy management challenges in multi-tenant residential buildings and found that landlords struggle with fair cost allocation while tenants lack visibility into their consumption, highlighting the need for systems that support both tenant-level and landlord-level monitoring. Bautista and Fernandez (2021) conducted a feasibility study on energy monitoring platforms in residential buildings and concluded that while monitoring technologies are technically viable, affordable

and user-friendly implementations are necessary for broader adoption. Reyes and Lim (2022) identified challenges in smart meter implementation, including infrastructure limitations, consumer acceptance barriers, regulatory gaps, and interoperability issues, reinforcing the importance of non-intrusive software approaches that do not depend on utility-level infrastructure. Torres and Aquino (2020) examined energy consumption behavior in Cagayan de Oro City and reported high tenant interest in monitoring technologies, particularly in boarding house environments. In addition, Mendoza and Cruz (2021) highlighted the significance of mobile-first application development for energy monitoring, noting user preference for cross-platform mobile solutions and intuitive visualization features. Ong and Sy (2019) further emphasized the advantages of cloud-based energy data management, including scalability, cost-effectiveness, and improved accessibility, while also noting connectivity and data privacy considerations. Collectively, these studies support the development of a software-based NILM system that integrates mobile visualization, cloud data management, and role-based monitoring tailored to Philippine residential environments.

Foreign Studies

NILM System Implementations

Foundational research on Non-Intrusive Load Monitoring (NILM) began with Hart (1992), who demonstrated that individual appliance consumption could be disaggregated from aggregated household energy measurements using steady-state and transient analysis. The study achieved approximately 90% accuracy in identifying major appliances and established the theoretical basis for NILM, while also highlighting limitations in distinguishing appliances with similar power consumption characteristics (Hart, 1992). Building on this work, Kelly and Knottenbelt (2015) developed the UK-DALE dataset and applied machine learning techniques, particularly Hidden Markov Models, achieving 85–95%



classification accuracy across more than 20 appliance types while enabling real-time processing and reproducible research through the NILMTK toolkit. European research by Bonfigli et al. (2018) demonstrated that residential energy monitoring systems can produce 15–20% energy savings by improving user awareness, with mobile interfaces significantly enhancing user satisfaction and engagement. More recent advances emphasize deep learning approaches, such as the bidirectional LSTM model proposed by Kaselimi et al. (2021), which achieved improved classification accuracy of up to 96% and better handling of complex appliance combinations while reducing computational requirements. Similarly, Nalmpantis and Vrakas (2020) explored time-series representations for NILM and identified effective feature extraction techniques that improve scalability and real-time performance. Beyond NILM algorithms, IoT-integrated energy monitoring research by Al-Ali et al. (2017) demonstrated that real-time monitoring combined with cloud analytics can reduce household energy consumption by 10–15% through behavioral changes and improved user engagement, particularly when mobile applications serve as the primary interface. Collectively, these studies highlight the evolution of NILM from rule-based approaches to advanced software-based analytics supported by cloud computing and mobile platforms, reinforcing the feasibility of developing scalable software-based residential energy monitoring systems.



Synthesis of Reviewed Literature and Studies

Based on the reviewed literature and studies, key themes and insights were identified.

1. Software NILM Has Become a Practical Solution

Recent studies show that NILM has evolved from theoretical research into practical software platforms capable of identifying appliance usage using aggregated energy datasets with high accuracy.

2. Cloud-Based Energy Monitoring Improves Accessibility and Scalability

Cloud computing enables real-time synchronization, scalable storage, and cross-platform access, allowing energy monitoring systems to operate efficiently without complex infrastructure.

3. Mobile-First Interfaces Enhance User Engagement

Literature consistently highlights that mobile applications providing real-time visualization, historical analytics, and alerts significantly improve energy awareness and user participation.

4. Role-Based Monitoring Is Important in Multi-Tenant Environments

Studies emphasize the need for systems that support both tenant-level and administrator-level monitoring to enable transparent reporting and fair energy cost allocation in rental properties.

5. Cost-Effective Software Solutions Support Wider Adoption

Research indicates that affordable, non-intrusive, software-driven monitoring platforms are more suitable for developing contexts where users prefer solutions that do not require complex installations.

6. Philippine Context Shows Strong Demand but Limited Implementation

Local studies reveal high interest in energy monitoring technologies, particularly in boarding houses, yet there is limited implementation of software-based NILM systems tailored to Philippine residential environments.



Research Gap

While numerous studies have explored Non-Intrusive Load Monitoring (NILM) technology and residential energy monitoring platforms, several gaps remain evident:

1. Limited NILM Implementation in the Philippine Residential Context

Most NILM research has been conducted in developed countries, resulting in limited studies that consider Philippine energy consumption patterns, cost structures, Internet connectivity conditions, and housing types such as boarding houses and rental units.

2. Few Systems Supporting Tenant–Landlord Energy Management

Existing monitoring platforms are primarily designed for single households. There is limited research on software systems that provide role-based monitoring where tenants can view individual consumption while landlords can access summarized unit-level data to support fair cost allocation.

3. Limited Mobile-First NILM Software Platforms

Although energy monitoring applications exist, many function only as dashboards rather than mobile-native NILM systems that support real-time analytics, appliance classification feedback, and cross-platform deployment.

4. Lack of Cost-Effective Cloud-Based Implementations

Many studies rely on complex infrastructure, creating a gap in research focused on affordable NILM software platforms that utilize cloud services, open-source tools, and scalable database solutions suitable for developing environments.

5. Insufficient Exploration of Real-Time Cloud Processing

There is limited research on NILM software systems that implement real-time data processing using serverless architectures, real-time synchronization, and cloud-native analytics pipelines.

6. Limited Integration of Modern Software Development Frameworks

Few studies demonstrate the combined use of modern mobile frameworks, cloud databases, and

software analytics tools in a single NILM platform, highlighting the need for integrated, user-centered energy monitoring solutions.

These gaps justify the development of the proposed software-based Non-Intrusive Load Monitoring (NILM) system tailored for Philippine residential environments.

This Study's Contribution

This research addresses the identified gaps by:

1. Developing a Philippine-Context NILM Software Platform

Designing a system tailored to residential environments, including boarding houses and rental properties.

2. Implementing Tenant–Landlord Monitoring Features

Providing role-based dashboards that support both individual consumption visibility and administrative reporting.

3. Adopting a Mobile-First Architecture

Creating a mobile-native NILM application that enables real-time visualization, analytics, and user interaction across platforms.

4. Utilizing Cost-Effective Cloud Technologies

Leveraging cloud databases, serverless processing, and open-source tools to create a scalable and affordable monitoring platform.

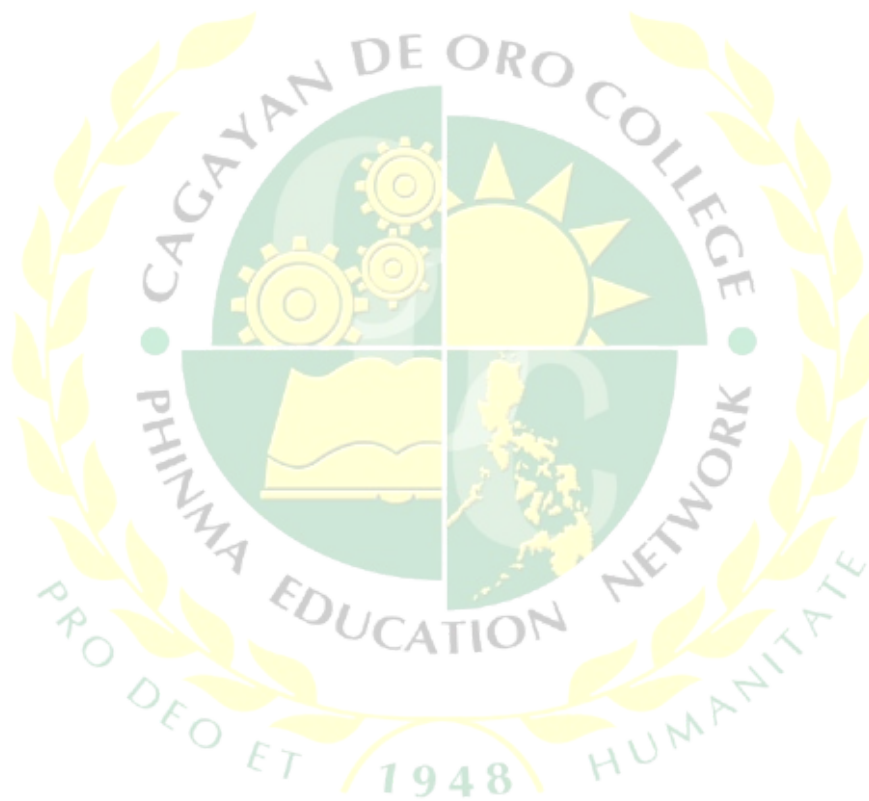
5. Enabling Real-Time Cloud Integration

Supporting continuous synchronization, automated analytics, and remote monitoring through cloud-based services.



6. Demonstrating an Integrated Software Technology Stack

Combining mobile development frameworks, cloud data management, and NILM analytics to provide a comprehensive residential energy monitoring solution.



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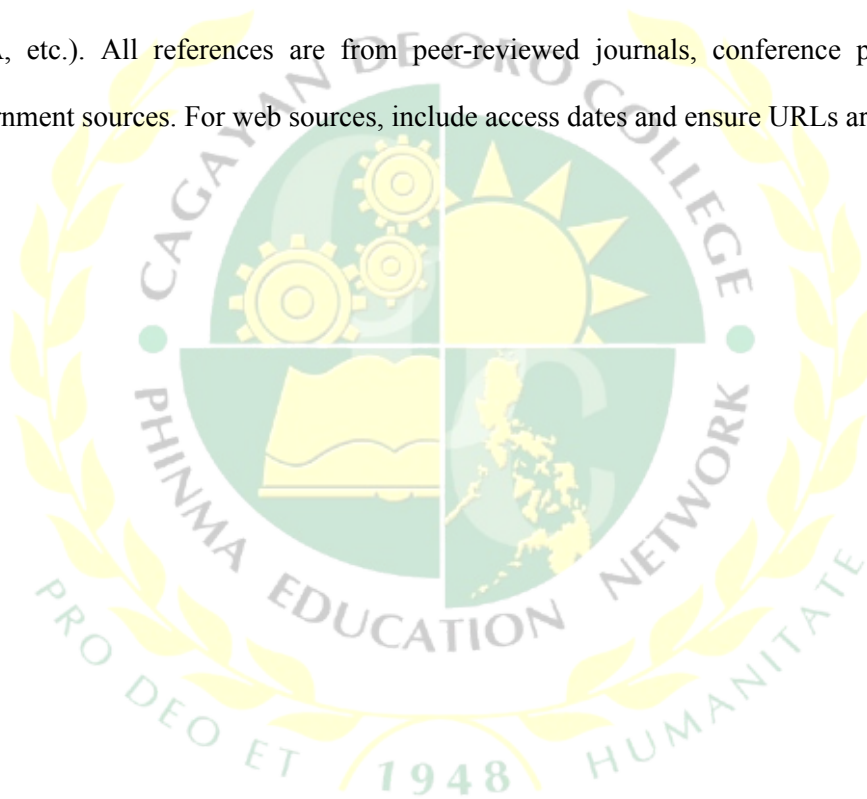
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CHAPTER 3 METHODOLOGY

Research Design

This study used descriptive–evaluative methods in a developmental research design. The study focused on the design, development, and evaluation of a software-based Non-Intrusive Load Monitoring (NILM) platform, particularly the mobile application and backend server services. The system utilized electrical data generated from external monitoring hardware developed in collaboration with the Department of Electrical Engineering, while the primary scope of the study remained on software processing, data management, and visualization features.

The descriptive–evaluative method was used to assess the system’s usability, accuracy, reliability, and performance efficiency through user and expert evaluation using recognized assessment instruments. Performance evaluations and accuracy tests were treated as part of system validation and assessment rather than experimental manipulation. Therefore, the study did not apply an experimental research design; instead, it emphasized the development of a functional software prototype and the evaluation of its effectiveness and quality using standardized evaluation tools.

Respondents and Sampling Technique

Purposive sampling was used to choose the study's respondents because they had traits that were pertinent to residential energy monitoring and system assessment. The following respondent groups participated in the study:

1. Tenants (Primary Users)

A total of thirty-five (35) tenants from selected boarding houses and rental units in Cagayan de Oro City served as primary users of the system. Participants were required to:



- regularly use household appliances;
- own an Android or iOS smartphone; and
- be willing to use and evaluate the system for at least four (4) weeks.

2. Landlords / Administrators

A total of seven (7) landlords/administrators participated in evaluating the landlord monitoring features. Participants were required to:

- manage at least three (3) rental units; and
- be willing to assess unit-level monitoring and reporting features.

3. Technical Evaluators

A total of eight (8) technical experts composed of Information Technology professionals and Electrical Engineers evaluated the system's technical quality and performance. They were selected based on experience in at least one of the following:

- IoT systems and embedded development;
- software and mobile application development; or
- energy monitoring and electrical measurement systems.

Research Locale

The study was conducted in selected boarding houses and rental properties in Cagayan de Oro City, Philippines. The locale was chosen due to the high population of student tenants, the dense concentration of rental housing, and the increasing demand for transparent energy monitoring systems in urban residential environments. According to the Philippine Statistics Authority, urban areas such as Cagayan de Oro City have experienced continuous population growth and expansion of rental housing to accommodate students and working residents (Philippine Statistics Authority, 2020). All participating sites utilized



standard residential electrical systems (220V, single-phase) and had stable internet connectivity to support real-time data synchronization and cloud-based monitoring.

Data Gathering Procedure

The data gathering procedure was conducted systematically to ensure the accuracy, reliability, and validity of the collected data for the development and evaluation of the Non-Intrusive Load Monitoring (NILM) system. The following steps were undertaken:

1. **Permission and Coordination** – Formal permission was secured from selected landlords and boarding house administrators in Cagayan de Oro City to install and test the NILM prototype in designated residential units. Participants were informed about the purpose of the study, and written consent was obtained prior to data collection.
2. **Requirements Gathering** – Interviews, surveys, and informal consultations were conducted with tenants and landlords to identify existing energy monitoring challenges, billing concerns, and desired system features. This phase helped define both functional and non-functional requirements of the system.
3. **Electrical Data Collection** – Electrical signals, including Voltage RMS and Current RMS, were collected using sensors connected to the ESP32 microcontroller. The system recorded real-time measurements such as Instantaneous Power (W), Apparent Power (VA), Power Factor (PF), and estimated Energy Consumption (kWh). The study primarily focused on the software processing and visualization of the collected electrical parameters, while the external hardware served only as the data acquisition source.



4. **System Development and Integration** – The gathered requirements and collected electrical parameters served as the basis for developing the mobile application and backend integration using Node.js and MySQL.
5. **System Testing and Validation** – The developed system underwent functional testing, integration testing, and appliance classification accuracy validation. Appliance identification results were compared with actual appliance usage to assess system performance.
6. **System Evaluation** – The system was evaluated by tenants, landlords, and technical experts using the System Usability Scale (SUS) and ISO/IEC 25010 software quality criteria to assess usability, reliability, performance efficiency, and overall system quality.
7. **Data Compilation and Statistical Analysis** – The evaluation results were collected, tabulated, and analyzed using appropriate statistical tools to determine overall system effectiveness, user satisfaction level, and technical performance metrics.



Research Instruments

The study used the following instruments to collect data and evaluate the system:

1. **System Requirements Questionnaire** – gathered user expectations, existing practices, and preferred features.
2. **System Usability Scale (SUS)** – measured perceived usability and user satisfaction.
3. **ISO/IEC 25010 Evaluation Checklist** – assessed software quality (functional suitability, reliability, usability, performance efficiency, security, maintainability).
4. **Accuracy Testing Protocol** – compared appliance classification results with actual appliance usage.
5. **Technical Performance Evaluation Form** – assessed response time, synchronization speed, and system stability.
6. **User Interview Guide** – collected qualitative feedback and improvement suggestions.
7. **System Logs and Analytics** – recorded transmission success rates, latency, and error frequency.

System Development Life Cycle (SDLC)

The study adopted the Modified Waterfall Model, a structured and sequential development process suitable for projects with defined software requirements and clear development phases.

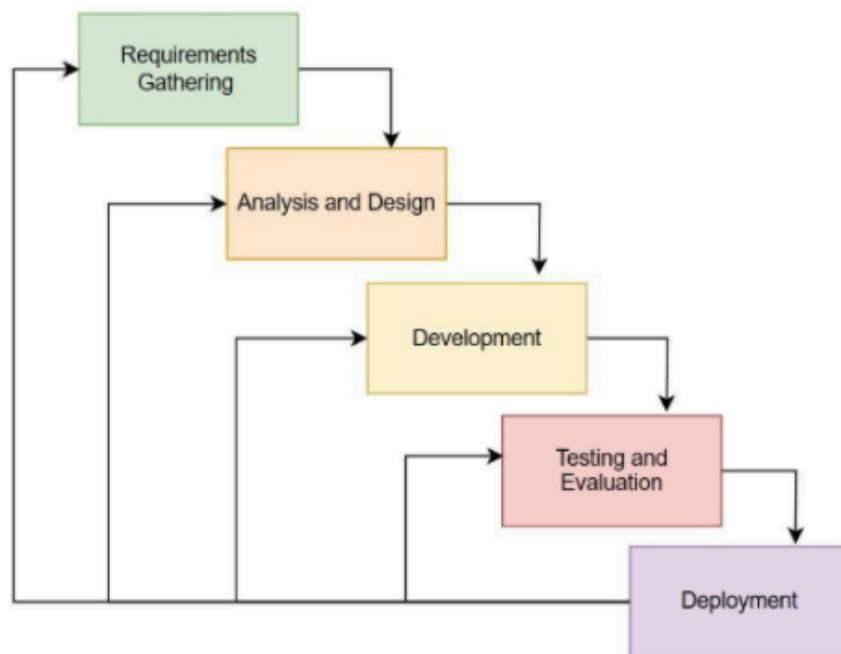


Figure 3.1 Modified Waterfall Model of SDLC

Modified Waterfall Model

This study adopted the Modified Waterfall Model in developing the software-based Non-Intrusive Load Monitoring (NILM) system. The model was selected because the project follows structured software development phases including requirements analysis, system design, application development, testing, deployment, and evaluation.

The system processes electrical consumption datasets generated from external monitoring devices developed in collaboration with the Department of Electrical Engineering. While hardware components provide data input, the primary focus of this study is on software processing, appliance classification, cloud integration, and mobile application visualization.

Although the model follows a sequential approach, limited feedback and refinement between phases were allowed, such as adjustments to data processing logic, classification algorithms, and user interface design. This approach ensures systematic development while supporting iterative software improvements.

Planning Phase

Defined the project scope, objectives, stakeholders, constraints, and success criteria. Activities included initial literature review, stakeholder consultations, selection of technology stack, risk assessment, and scheduling.

Requirements Gathering Phase

Collected user requirements through surveys and interviews. Functional and non-functional requirements were identified, including real-time monitoring, appliance classification, mobile accessibility, accuracy, security, and performance expectations.

Analysis and Design Phase

The Analysis and Design Phase involved translating the gathered requirements into structured system models and software architecture for the proposed Non-Intrusive Load Monitoring (NILM) system. This phase ensured that the software components were clearly defined before development.

The phase included the following:

- **Conceptual Design** – Developed context-level diagrams illustrating the interaction between tenants, landlords, external monitoring devices, and the NILM software system. This showed how electrical consumption data flows into the software for processing and visualization.



- **Use Case Diagrams** – Illustrated key system operations such as user registration, real-time energy monitoring, appliance classification visualization, report generation, and role-based access control for tenants and landlords.
- **Database Design** – Created database models defining data structures for user accounts, energy consumption records, appliance events, and system logs. Logical schema design ensured organized storage and efficient retrieval of monitoring data.
- **Interface Design** – Designed mobile application layouts, dashboards, and visualization components to provide clear and user-friendly presentation of real-time energy usage and appliance insights.
- **Process Flowcharts** – Mapped existing manual energy monitoring practices and the proposed automated workflow to illustrate improvements in data processing, reporting, and accessibility.

This phase provided the foundation for developing a scalable, maintainable, and user-centered NILM software system.

Development Phase

This phase involved the actual construction and integration of the proposed Software-Based Non-Intrusive Load Monitoring (NILM) System based on the finalized system requirements and design specifications. The software development process utilized electrical parameter data obtained from external monitoring devices developed in collaboration with the Department of Electrical Engineering. While the hardware components served as the data acquisition source, the primary focus of the study remained on software processing, cloud integration, and mobile application visualization.

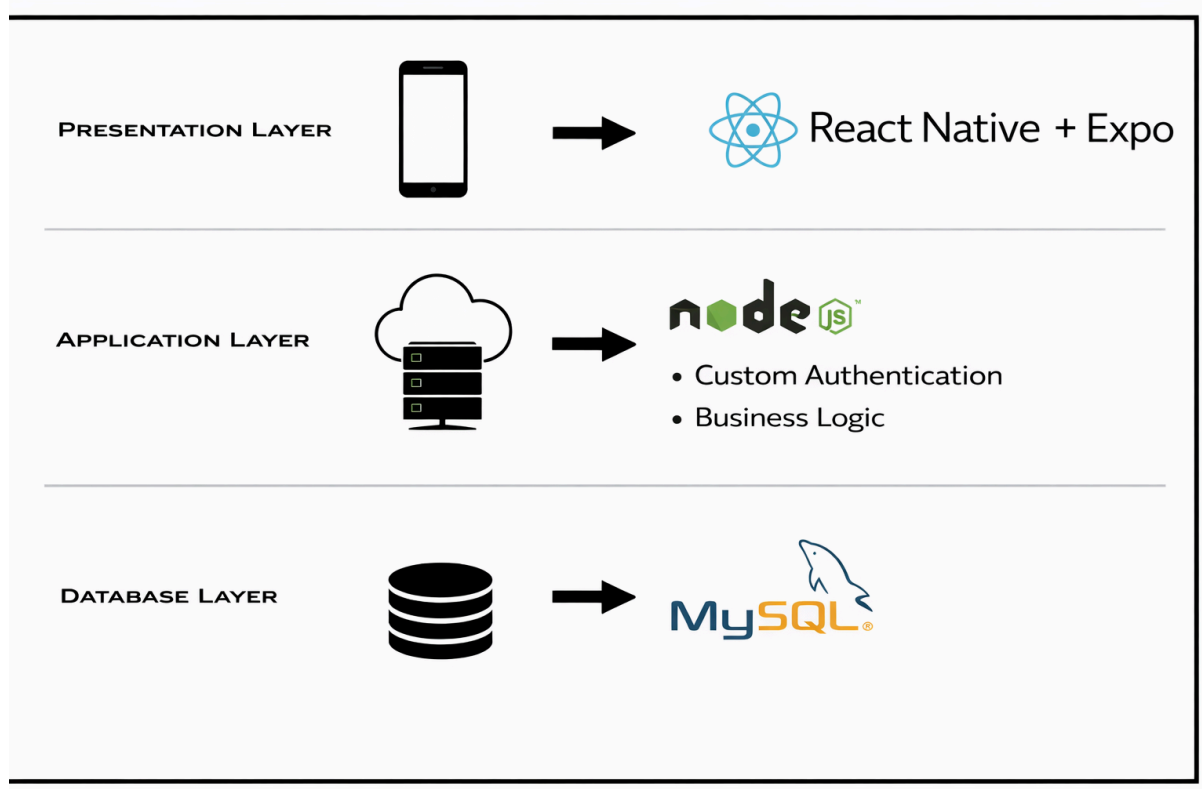


Figure 3.2 Three-Layers Software Architecture of the Proposed NILM System

The system adopted a three-layer architecture composed of the presentation layer, application layer, and database layer.

Presentation Layer The presentation layer consists of a cross-platform mobile application developed using React Native + Expo. This layer handles user interaction, data visualization, real-time monitoring dashboards, and role-based access for tenants and landlords.

Application Layer The application layer implements backend services using Node.js, which manages custom authentication and core business logic. This includes processing of energy data, validation of incoming readings, generation of summaries, and execution of rule-based conditions for alerts and notifications.



Database Layer The database layer uses MySQL for structured storage and retrieval of system records. This layer stores user accounts, unit-level monitoring data, appliance activity logs, and historical consumption reports to support real-time monitoring and analytics.

All components were integrated to enable seamless data processing, secure authentication, and efficient retrieval of monitoring data for both tenant and landlord dashboards.

Testing and Evaluation Phase

System testing and evaluation were conducted to ensure that the developed **software-based Non-Intrusive Load Monitoring (NILM) system** functioned accurately, reliably, and according to the defined requirements. This phase focused on validating the software processing, real-time monitoring features, cloud synchronization, and overall usability of the system.

System testing included:

- **Functionality Testing** – Verified that core software features operated correctly, including user authentication, real-time energy monitoring dashboards, appliance classification visualization, and report generation.
- **Integration Testing** – Ensured successful mobile application, Node.js backend server, and MySQL database, and external monitoring devices providing electrical consumption data.
- **Scenario-Based Testing** – Simulated tenant and landlord interactions such as incorrect login attempts, missing data entries, delayed synchronization, and appliance detection events to evaluate system behavior under different conditions.

- **Evaluation** – Conducted with selected respondents using **ISO/IEC 25010** and the **System Usability Scale (SUS)** to assess software quality, usability, reliability, performance efficiency, and security.

- **Reliability Testing** – Statistical measures such as **Cronbach's Alpha** were used to verify the internal consistency of survey responses and ensure reliability of evaluation results.

Baseline comparison between existing manual electricity monitoring practices and the developed NILM software measured improvements in data accessibility, monitoring accuracy, transparency, and user experience.

Deployment Phase

After successful testing and evaluation, the developed **software-based Non-Intrusive Load Monitoring (NILM) system** was prepared for deployment in selected residential units. The deployment phase involved configuring the mobile application, cloud database, and user access settings to support real-time energy monitoring.

The software platform was connected to external monitoring devices that provided electrical consumption data, enabling seamless synchronization between data acquisition and system visualization. Tenants were oriented on how to use the mobile application to monitor real-time energy usage, view appliance insights, and access consumption reports. Landlords were guided on accessing unit-level summaries and monitoring dashboards.

Deployment ensured that the NILM software operated in a real-world environment, providing improved visibility of electricity consumption, enhanced reporting capabilities, and accessible monitoring features for both tenants and administrators. This phase confirmed the system's readiness for practical use and long-term implementation.

Tools, Technologies, and Technical Environment

The development of the proposed software-based **Non-Intrusive Load Monitoring (NILM) system** utilized various tools, technologies, and technical platforms to support mobile application development, backend integration, and real-time data visualization.

- **Programming Languages: JavaScript, TypeScript**

Used for mobile application development and backend server logic.

- **Frameworks/Libraries: React Native, Expo**

Enabled cross-platform mobile application development with responsive user interfaces and real-time monitoring dashboards.

- **Backend Platform: Node.js**

Provided API development, server-side business logic, data validation, and processing of energy monitoring records.

- **Database: MySQL**

Used for structured storage and retrieval of user accounts, unit-level monitoring data, appliance activity logs, and historical energy consumption records.

- **Authentication: Custom Authentication**

Implemented role-based access control for tenants and landlords to ensure secure system access and data privacy.

- **Development Tools: Visual Studio Code, GitHub**

Used for coding, version control, collaboration, and project management.

- **Operating System: Windows 10**

Served as the primary development environment for building and testing the application.



- **Mobile Platform: Android and iOS**

Supported deployment of the NILM mobile application for tenants and landlords.

- **Web Browser: Google Chrome**

Used for system testing, API monitoring, and backend configuration. These tools were selected due to their compatibility, scalability, availability, and suitability for developing a backend-integrated mobile monitoring software system.

Ethical Considerations

The study adhered to the provisions of the **Data Privacy Act of 2012 (RA 10173)** to ensure the protection of participant information throughout the development and evaluation of the **software-based Non-Intrusive Load Monitoring (NILM) system**. Informed consent was obtained from all participants, and they were informed of the study's purpose, procedures, and intended use of collected data.

Data collection and handling followed the principles of confidentiality, data minimization, and secure storage. Only information necessary for system functionality and evaluation was collected. The system implemented **role-based access control (RBAC)** to restrict data visibility according to user roles and prevent unauthorized access.

Security measures, including authentication, controlled cloud database access, and system activity logging, were applied to maintain data integrity and accountability. Participants were given the right to withdraw at any time, and all collected information was handled in accordance with established ethical research standards.



Appendix A

Appendix A: Mockup Design

Login page

9:41

Smart Energy Monitor

Track your energy, save your budget

Email Address

Password

☐ Remember me [Forgot Password?](#)

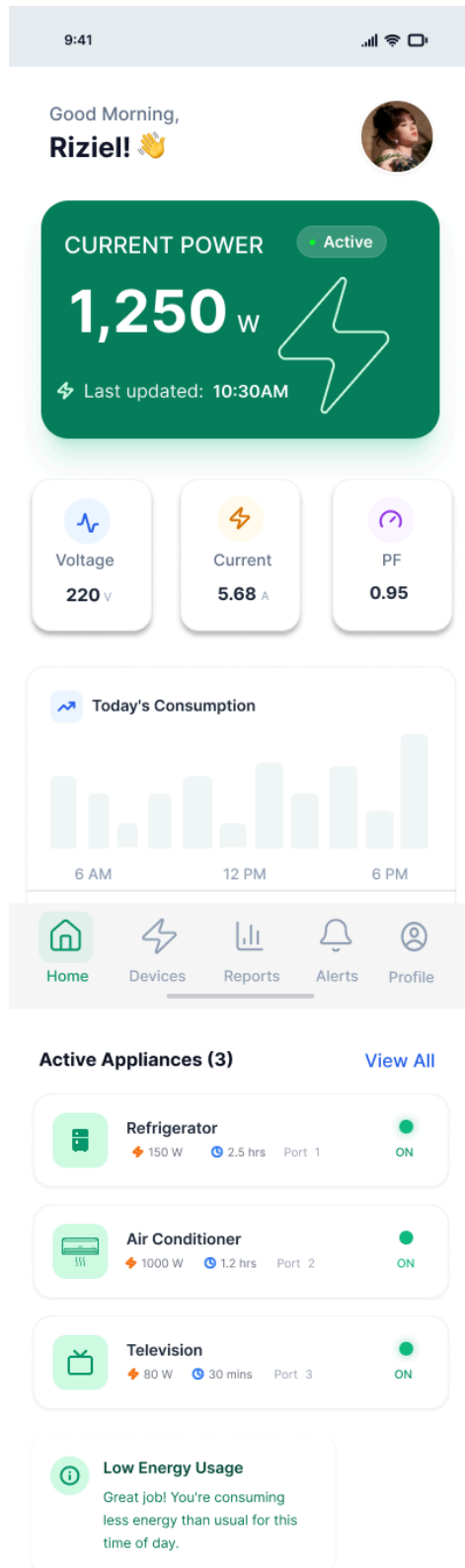
Sign In →

OR

Continue with Google

Don't have an account? [Sign Up](#)

By signing in, you agree to our [Terms & Privacy Policy](#)
Home page (tenants)



Home page (landlord)



9:41

Property Overview

Sunset Apartments

3/4 Online

Total Building Load

5,250 W

Today's Total

38.5 kWh

Est. Revenue

₱481.25

4

Total Units

3

3

Online

+12%

vs Last Week

Units (5)

Unit 1 - Main Meter John Doe	₱ 1,881 Cost	1250 W 10.8 kWh
Unit 2 - Meter Jane Smith	₱ 1,504 Cost	1500 W 12.5 kWh
Unit 3 - Meter Bob Johnson	₱2,253 Cost	2500 W 15.2 kWh
Unit 4 - Meter Vacant	₱ 2,073 Cost	0 W 0 kWh

Home

Tenant

Reports

Alerts

Profile

Unit 1

Unit 2

Unit 3

Unit 4

View Reports

Generate Bills



Appendix B: Letter of Intent

CAGAYAN DE ORO COLLEGE PHINMA EDUCATION

February 13, 2026
College of Engineering
Electrical Engineering Department
PHINMA COC

Dear Sir/Madam,

Greetings!

We, the undersigned third-year Bachelor of Science in Information Technology students from PHINMA Cagayan de Oro College, namely: ARGATE KENT, BERSANO RIZIEL, DALIT REYMARK, EMBORGO NEIL JOEBERT, GAROL MIKHAELA, NOVECIO GIL JOHN, respectfully write to request your permission to conduct a research study and system development project within your esteemed [Office/Business].

This initiative is in partial fulfillment of the requirements for our course *Quantitative Methods leading to Capstone Research 1*, which may ultimately lead to the development of a computer-based system tailored to the needs of your organization. As part of our academic undertaking, we seek to conduct field research and system testing within your premises.

We assure you that all information and data collected during the research will be treated with the utmost confidentiality and will be used solely for academic purposes.

We sincerely hope for your favorable consideration and kind support of our proposal. Your approval will greatly contribute to the success of our academic requirements and to our growth as aspiring IT professionals.

Thank you very much, and we look forward to your positive response.

Respectfully yours,

ARGATE, KENT
BERSANO, RIZIEL
DALIT, REYMARK
BSIT 3rd Year Students
PHINMA Cagayan de Oro College

EMBORGO, NEIL JOEBERT
GAROL, MIKHAELA
NOVECIO, GIL JOHN

Noted by:

ALEXIS V. LAROSA, PhD
Instructor/Adviser

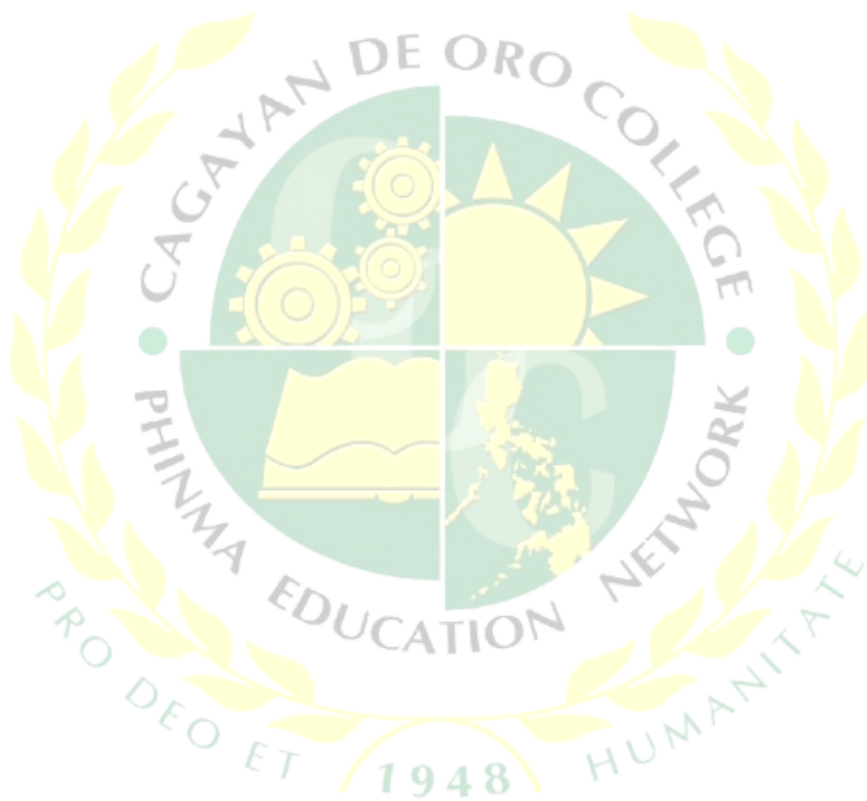
Conforme:

I hereby acknowledge and approve the request of the above-mentioned students to conduct their research study and system project within our office/business. I understand that all data gathered will be kept confidential and used strictly for academic purposes.

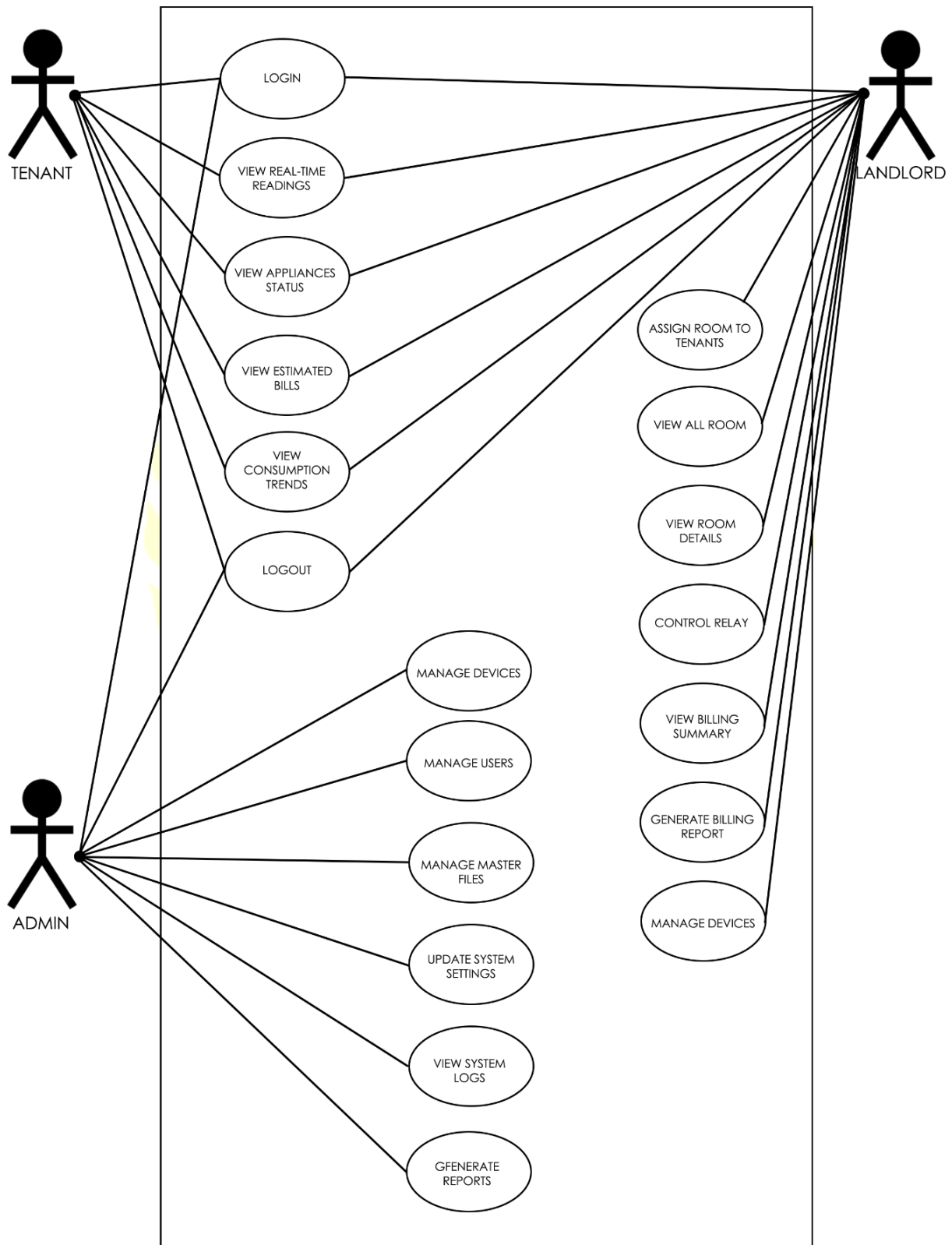
Approved by:
BOLAN, MARK ANTHONY R.

Appendix C: Gantt Chart of Activities

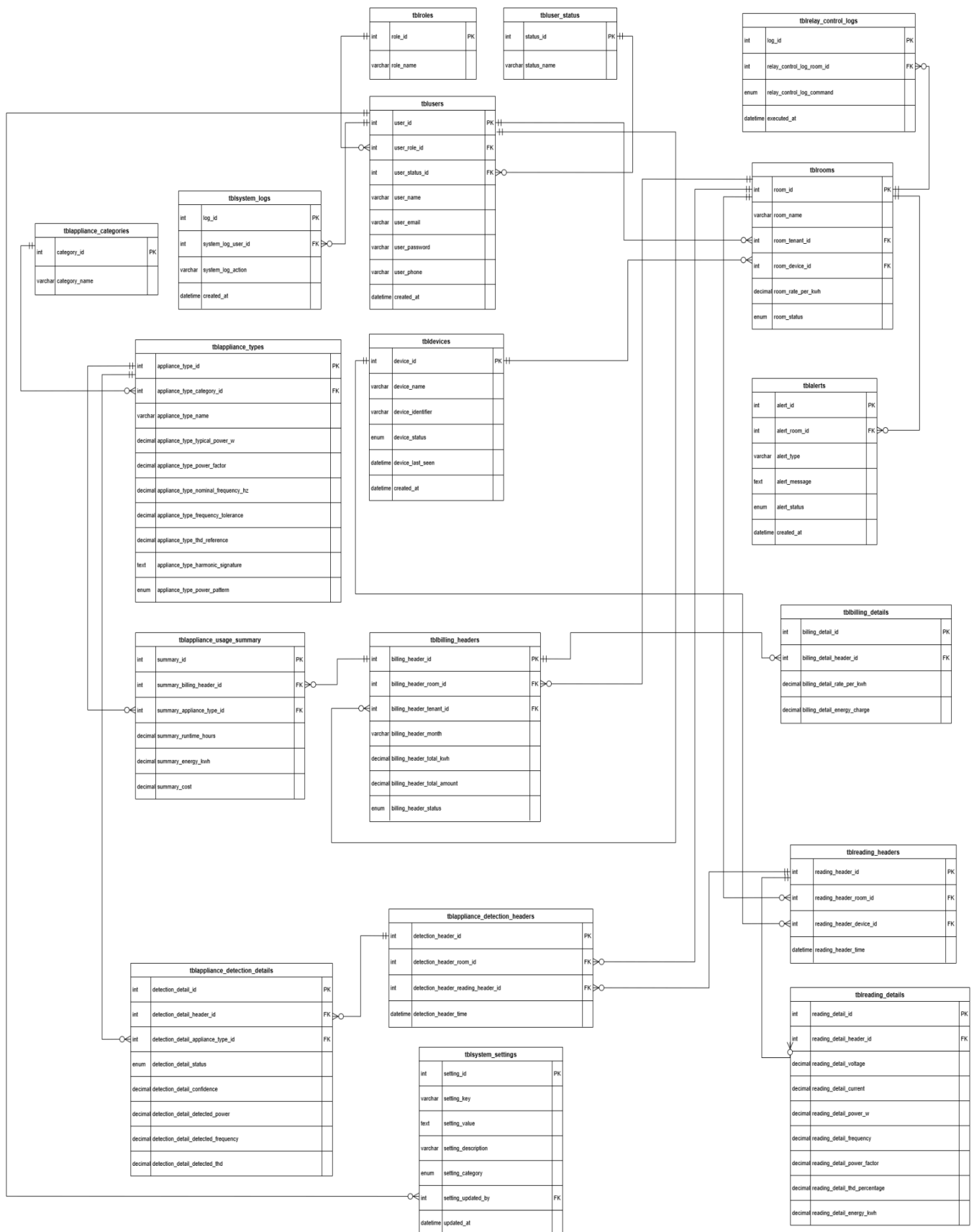
A	B	C	D	E	F	G	H	I
Gantt-chart	January				February			
	week1	week2	week3	week4	week1	week2	week3	week4
client requirement gathering								
chapters 1								
system design & planning								
chapter 2 & 3								
UI/UX								
Login, Registration, Forgot Password & Admin Dashboard Completed								



Appendix D: Use Case Diagram

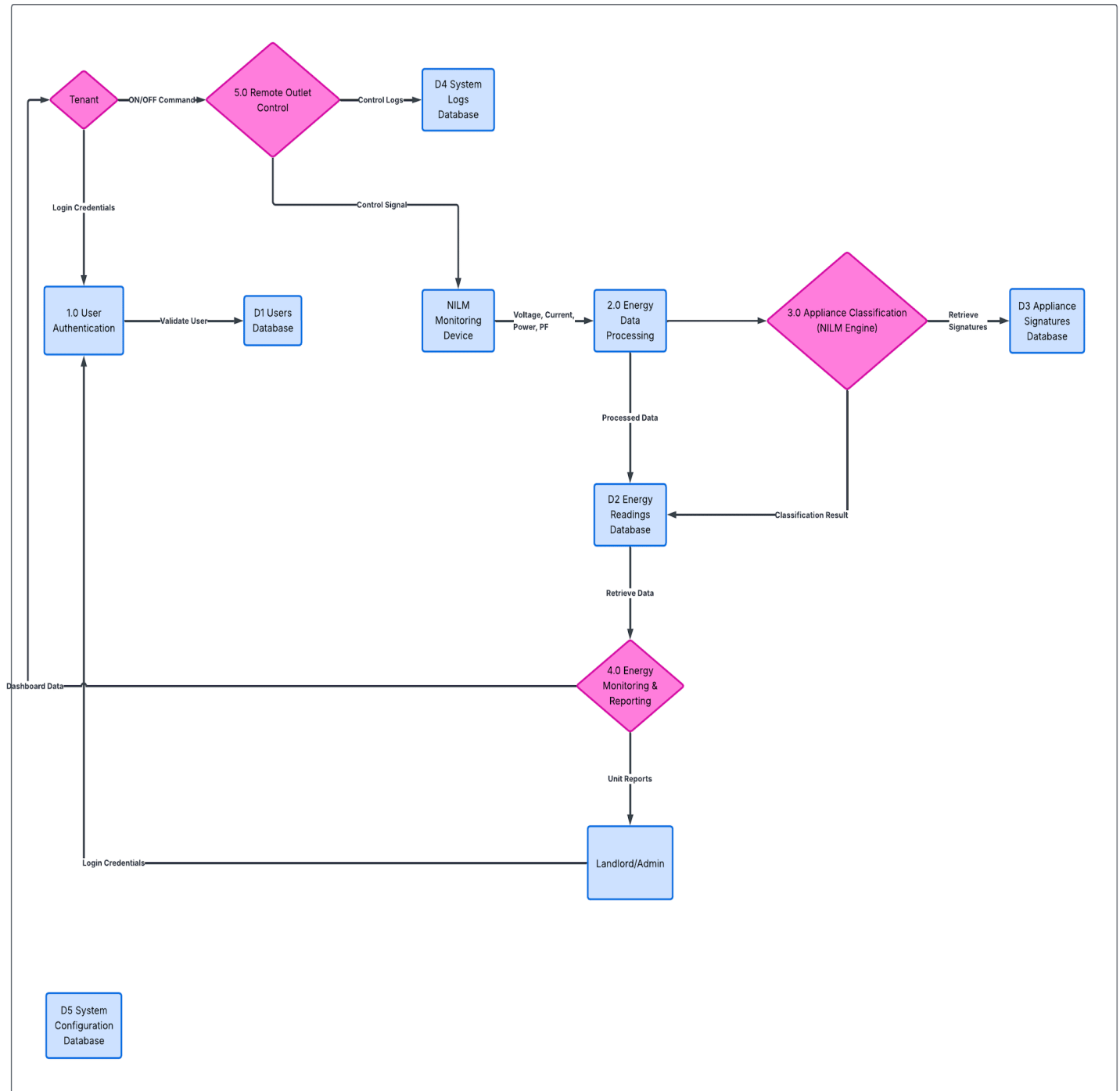


Appendix E: Logical Entity Relationship Diagram (ERD)

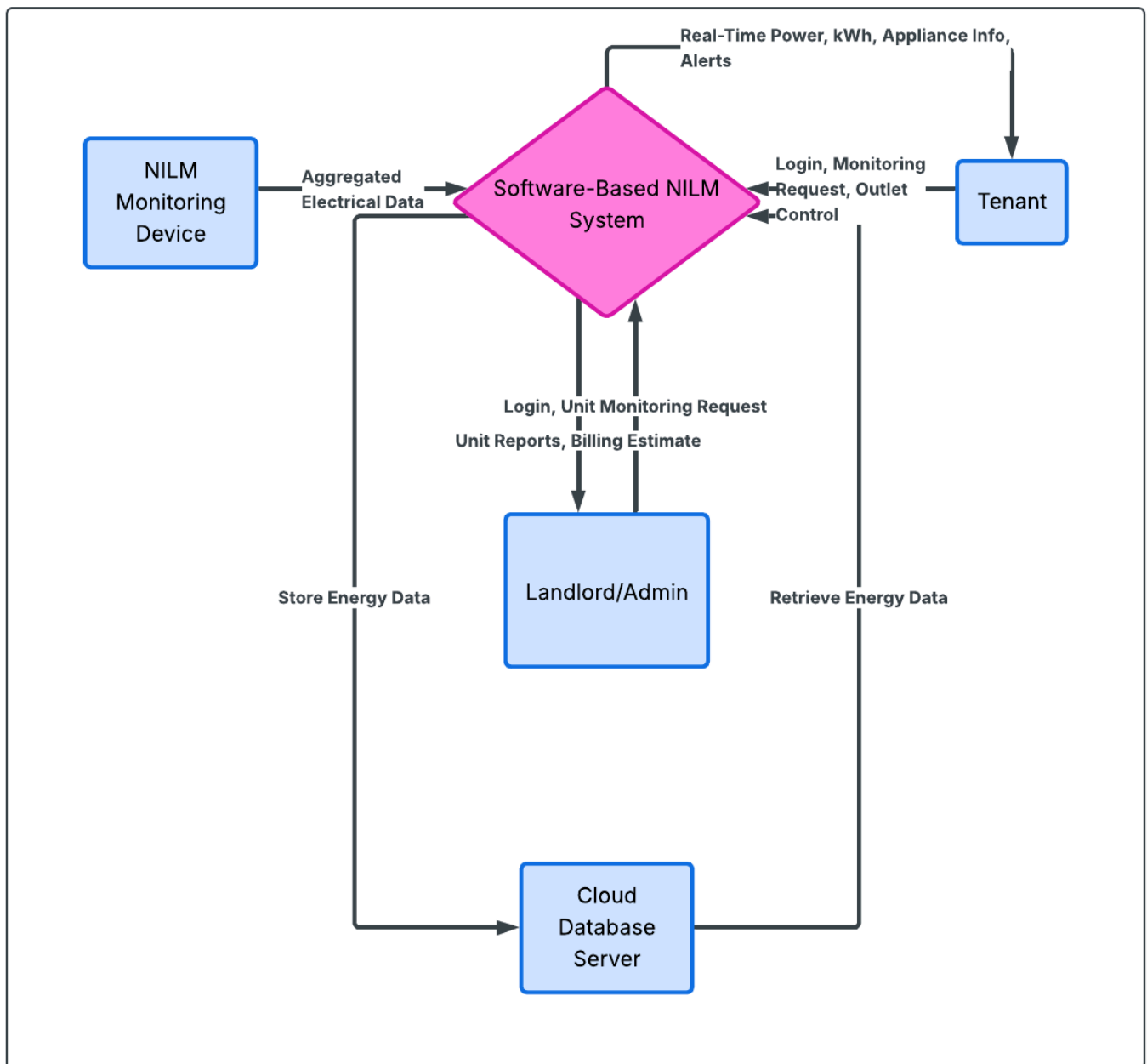


Appendix F: Data Flow Diagram

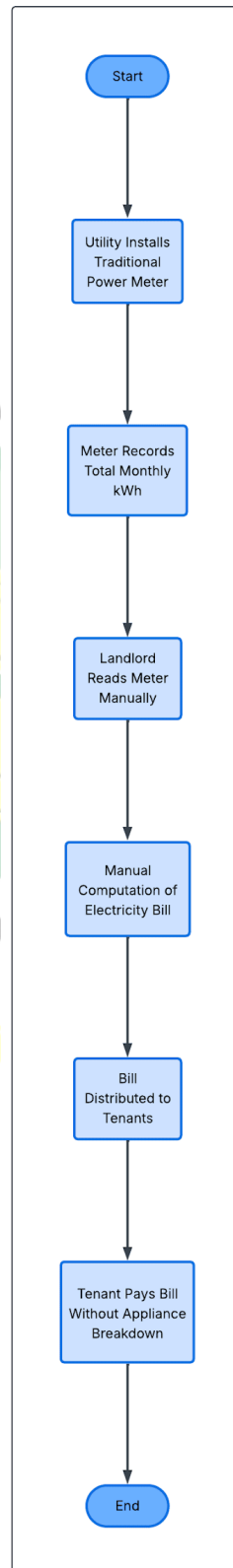
Data Flow Diagram



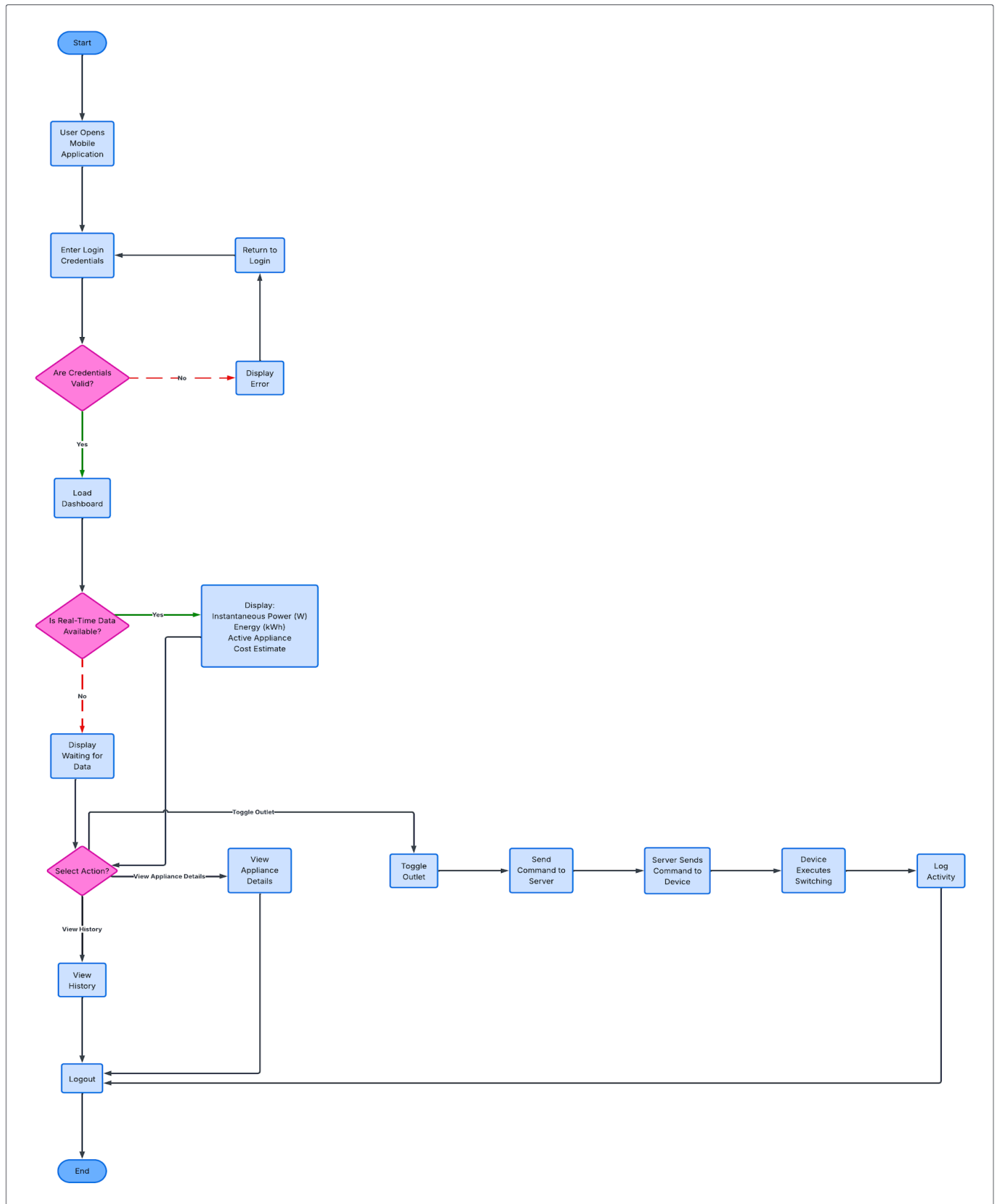
Appendix G: Context Diagram



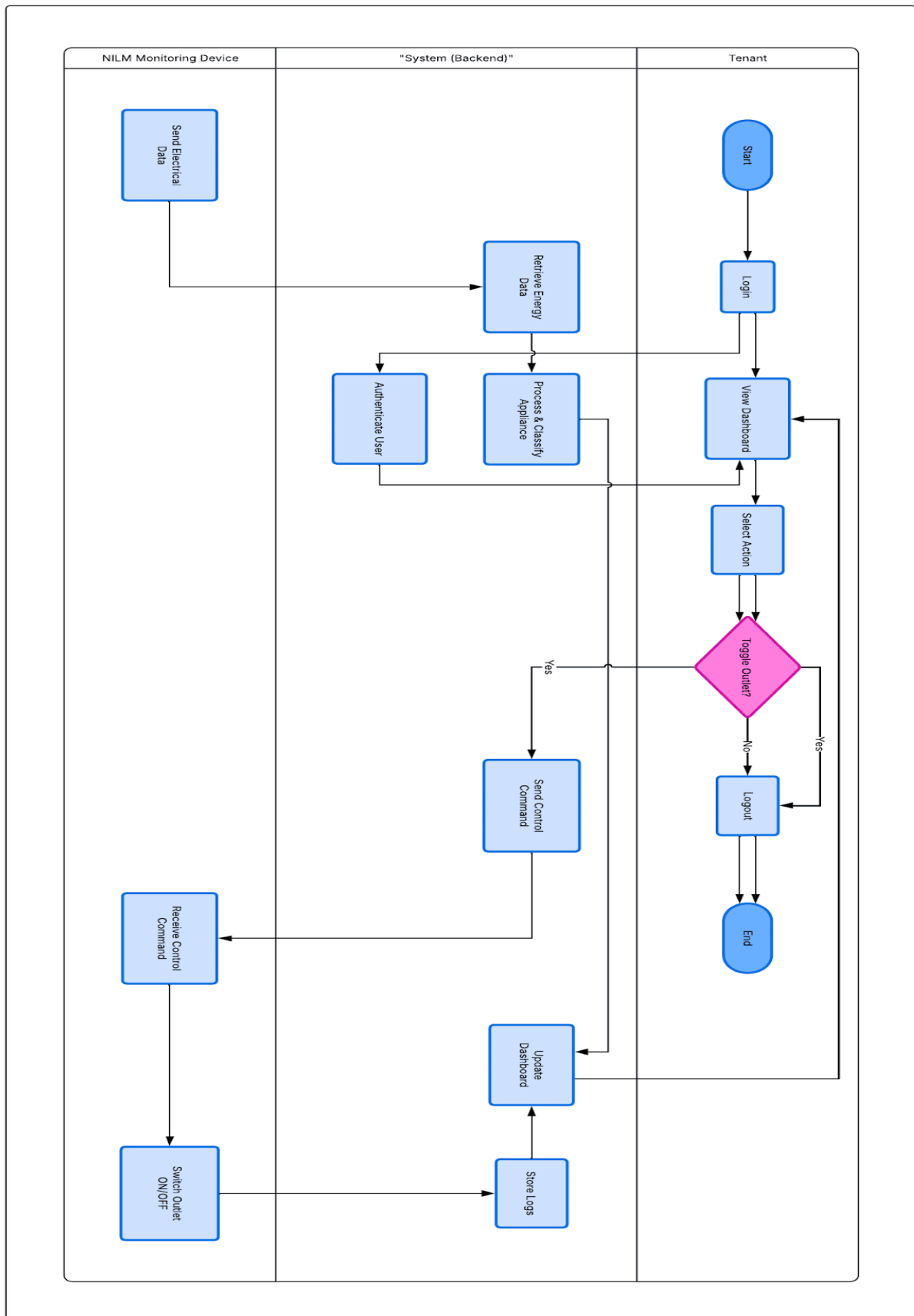
Appendix H: Manual Process Flowchart



Appendix I: Proposed System Flowchart



Appendix J: System Activity Diagram



Appendix K: Data Dictionary

Table: tblroles

Field Name	Data Type	Length	Key / Constraints	Description
role_id	INT	11	PK, AUTO_INCREMENT	Unique role identifier
role_name	VARCHAR	50	UNIQUE, NOT NULL	Name of user role

Table: tbluser_status

Field Name	Data Type	Length	Key / Constraints	Description
status_id	INT	11	PK, AUTO_INCREMENT	Unique status identifier
status_name	VARCHAR	50	UNIQUE, NOT NULL	Account status (active, inactive, suspended)

Table: tblusers

Field Name	Data Type	Length	Key / Constraints	Description
user_id	INT	11	PK, AUTO_INCREMENT	Unique user identifier
user_role_id	INT	11	FK → tblroles(role_id), NOT NULL	User role reference
user_status_id	INT	11	FK → tbluser_status(status_id), NOT NULL	User status reference
user_name	VARCHAR	100	NOT NULL	Full name of user
user_email	VARCHAR	100	UNIQUE, NOT NULL	Email address
user_password	VARCHAR	255	NOT NULL	Encrypted password
user_phone	VARCHAR	20	NULL	Contact number
created_at	DATETIME	—	DEFAULT CURRENT_TIMESTAMP	Account creation date

Table: tblappliance_categories

Field Name	Data Type	Length	Key / Constraints	Description
category_id	INT	11	PK, AUTO_INCREMENT	Appliance category ID
category_name	VARCHAR	50	UNIQUE, NOT NULL	Appliance category name

**Table: tblappliance_types**

Field Name	Data Type	Length	Key / Constraints	Description
appliance_type_id	INT	11	PK, AUTO_INCREMENT	Appliance type ID
appliance_type_category_id	INT	11	FK → tblappliance_categories(category_id), NOT NULL	Category reference
appliance_type_name	VARCHAR	100	NULL	Appliance name
appliance_type_typical_power_w	DECIMAL	10,2	NULL	Typical power (Watts)
appliance_type_power_factor	DECIMAL	5,2	NULL	Typical power factor
appliance_type_nominal_frequency_hz	DECIMAL	5,2	DEFAULT 60.00	Nominal frequency (Hz)
appliance_type_frequency_tolerance	DECIMAL	5,2	DEFAULT 0.50	Frequency tolerance
appliance_type_thd_reference	DECIMAL	5,2	NULL	Reference THD (%)
appliance_type_harmonic_signature	TEXT	—	NULL	Harmonic signature (JSON format)
appliance_type_power_pattern	ENUM	constant',cyclic',variable'	DEFAULT 'constant'	Power behavior pattern

Table: tbldevices

Field Name	Data Type	Length	Key / Constraints	Description
device_id	INT	11	PK, AUTO_INCREMENT	Device ID
device_name	VARCHAR	100	NULL	Device name
device_identifier	VARCHAR	100	UNIQUE	Unique device code
device_status	ENUM	online','offline'	DEFAULT 'online'	Device status
device_last_seen	DATETIME	—	NULL	Last communication timestamp



created_at	DATETIME	—	DEFAULT CURRENT_TIMESTAMP	Date registered
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Table: tblrooms

Field Name	Data Type	Length / Values	Key / Constraints	Description
room_id	INT	11	PK, AUTO_INCREMENT	Room ID
room_name	VARCHAR	100	NULL	Room name
room_tenant_id	INT	11	FK → tblusers(user_id)	Assigned tenant
room_device_id	INT	11	FK → tbldevices(device_id)	Monitoring device reference
room_rate_per_kwh	DECIMAL	10,2	DEFAULT 12.00	Electricity rate per kWh
room_status	ENUM	available',occupied'	DEFAULT 'occupied'	Occupancy status

Table: tblreading_headers

Field Name	Data Type	Length / Values	Key / Constraints	Description
reading_header_id	INT	11	PK, AUTO_INCREMENT	Reading session ID
reading_header_room_id	INT	11	FK → tblrooms(room_id)	Room reference
reading_header_device_id	INT	11	FK → tbldevices(device_id)	Device reference
reading_header_time	DATETIME	—	NULL	Reading timestamp

Table: tblreading_details

Field Name	Data Type	Length	Key / Constraints	Description
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reading_detail_id	INT	11	PK, AUTO_INCREMENT	Reading detail ID
reading_detail_header_id	INT	11	FK → tblreading_headers(reading_header_id)	Header reference
reading_detail_voltage	DECIMAL	10,2	NULL	Voltage (V)
reading_detail_current	DECIMAL	10,3	NULL	Current (A)
reading_detail_power_w	DECIMAL	10,2	NULL	Power (Watts)
reading_detail_frequency	DECIMAL	10,2	NULL	Frequency (Hz)
reading_detail_power_factor	DECIMAL	5,2	NULL	Power factor
reading_detail_thd_percentage	DECIMAL	5,2	NULL	THD (%)
reading_detail_energy_kwh	DECIMAL	10,4	NULL	Energy consumption (kWh)

Table: tblappliance_detection_headers

Field Name	Data Type	Length	Key / Constraints	Description
detection_header_id	INT	11	PK, AUTO_INCREMENT	Detection session ID
detection_header_room_id	INT	11	FK → tblrooms(room_id)	Room reference
detection_header_reading_header_id	INT	11	FK → tblreading_headers(r eading_header_id)	Reading reference
detection_header_time	DATETIME	—	NULL	Detection timestamp

Table: tblappliance_detection_details

Field Name	Data Type	Length / Values	Key / Constraints	Description
detection_detail_id	INT	11	PK, AUTO_INCREMENT	Detection detail ID
detection_detail_header_id	INT	11	FK → tblappliance_detection_h eaders(detection_header_id)	Header reference



detection_detail_appliance_type_id	INT	11	FK → tblappliance_types(appliance_type_id)	Appliance reference
detection_detail_status	ENUM	ON','OFF'	NULL	Appliance state
detection_detail_confidence	DECIMAL	5,2	NULL	Confidence score
detection_detail_detected_power	DECIMAL	10,2	NULL	Detected power
detection_detail_detected_frequency	DECIMAL	10,2	NULL	Detected frequency
detection_detail_detected_thd	DECIMAL	5,2	NULL	Detected THD

Table: tblbilling_headers

Field Name	Data Type	Length / Values	Key / Constraints	Description
billing_header_id	INT	11	PK, AUTO_INCREMENT	Billing ID
billing_header_room_id	INT	11	FK → tblrooms(room_id)	Room reference
billing_header_tenant_id	INT	11	FK → tblusers(user_id)	Tenant reference
billing_header_month	VARCHAR	7	NULL	Billing month (YYYY-MM)
billing_header_total_kwh	DECIMAL	10,2	NULL	Total kWh consumed
billing_header_total_amount	DECIMAL	10,2	NULL	Total billing amount
billing_header_status	ENUM	pending','paid'	DEFAULT 'pending'	Payment status

Table: tblbilling_details

Field Name	Data Type	Length / Values	Key / Constraints	Description
billing_detail_id	INT	11	PK, AUTO_INCREMENT	Billing detail ID



billing_detail_header_id	INT	11	FK → tblbilling_headers(billing_header_id)	Header reference
billing_detail_rate_per_kwh	DECIMAL	10,2	NULL	Electricity rate
billing_detail_energy_charge	DECIMAL	10,2	NULL	Energy charge amount

Table: tblappliance_usage_summary

Field Name	Data Type	Length / Values	Key / Constraints	Description
summary_id	INT	11	PK, AUTO_INCREMENT	Summary ID
summary_billing_header_id	INT	11	FK → tblbilling_headers(billing_header_id)	Billing reference
summary_appliance_type_id	INT	11	FK → tblappliance_types(appliance_type_id)	Appliance reference
summary_runtime_hours	DECIMAL	10,2	NULL	Runtime hours
summary_energy_kwh	DECIMAL	10,2	NULL	Energy consumed
summary_cost	DECIMAL	10,2	NULL	Computed cost

Table: tblalerts

Field Name	Data Type	Length / Values	Key / Constraints	Description
alert_id	INT	11	PK, AUTO_INCREMENT	Alert ID
alert_room_id	INT	11	FK → tblrooms(room_id)	Room reference
alert_type	VARCHAR	50	NULL	Alert type
alert_message	TEXT	—	NULL	Alert message
alert_status	ENUM	new', resolved'	DEFAULT 'new'	Alert status
created_at	DATETIME	—	DEFAULT CURRENT_TIMESTAMP	Creation timestamp

**Table: tblrelay_control_logs**

Field Name	Data Type	Length / Values	Key / Constraints	Description
log_id	INT	11	PK, AUTO_INCREMENT	Log ID
relay_control_log_room_id	INT	11	FK → tblrooms(room_id)	Room reference
relay_control_log_command	ENUM	ON','OFF'	NULL	Relay command
executed_at	DATETIME	—	DEFAULT CURRENT_TIMESTAMP	Execution time

Table: tblsystem_settings

Field Name	Data Type	Length / Values	Key / Constraints	Description
setting_id	INT	11	PK, AUTO_INCREMENT	Setting ID
setting_key	VARCHAR	100	UNIQUE, NOT NULL	Setting key
setting_value	TEXT	—	NULL	Setting value
setting_description	VARCHAR	255	NULL	Setting description
setting_category	ENUM	'billing','detection','alerts','general'	DEFAULT 'general'	Setting category
setting_updated_by	INT	11	FK → tblusers(user_id)	Updated by
updated_at	DATETIME	—	ON UPDATE CURRENT_TIMESTAMP	Last update time

Table: tblsystem_logs

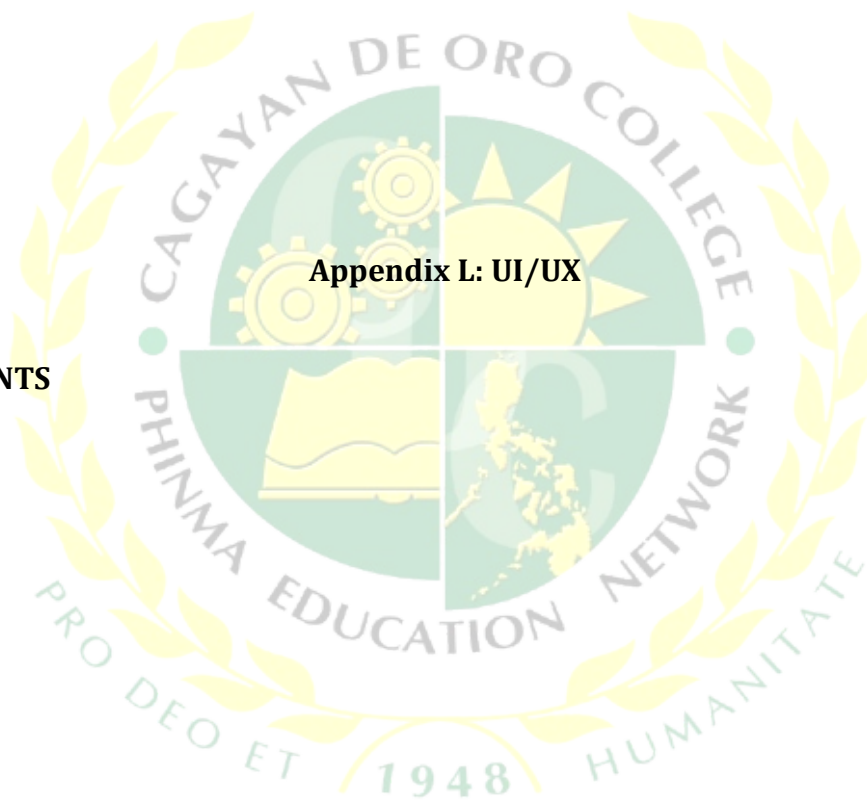
Field Name	Data Type	Length / Values	Key / Constraints	Description
log_id	INT	11	PK, AUTO_INCREMENT	Log ID
system_log_user_id	INT	11	FK → tblusers(user_id)	User reference
system_log_action	VARCHAR	255	NULL	Action performed

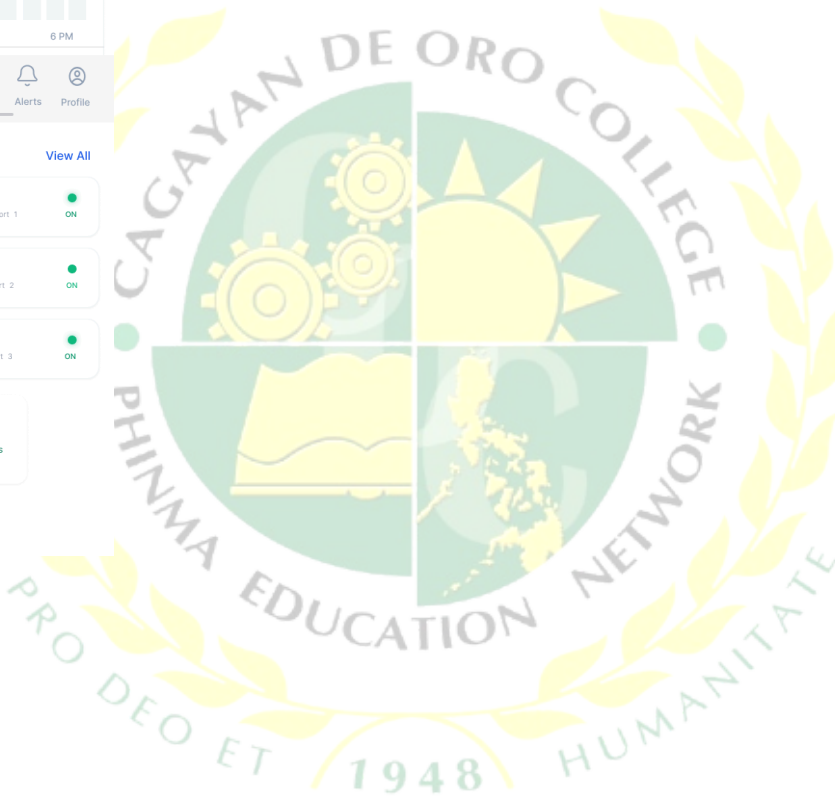
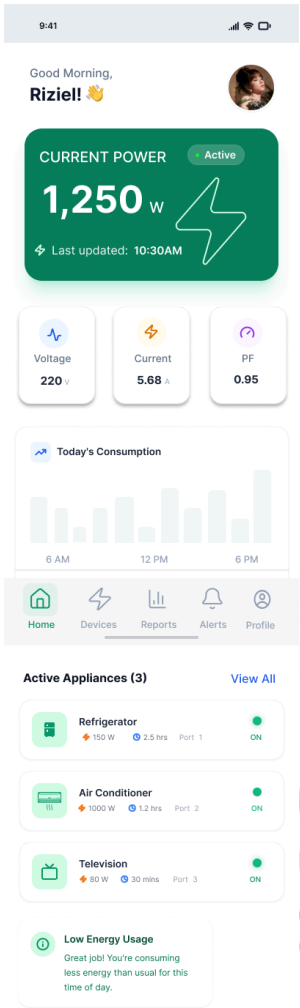


created_at	DATETIME	—	DEFAULT CURRENT_TIMESTAMP	Timestamp
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POV TENANTS

DASHBOARD





Appliances



9:41

<

Appliances

Search appliances...

All Devices

All Status

5 appliances • 3 active

Refrigerator

150 W

5.6

40.1

Current

kWh today

kWh Month

ON

Air Conditioner

1000 W

12.0

50.2

Current

kWh today

kWh Month

ON

Television

0 W

0.8

25.3

Current

kWh today

kWh Month

OFF

Living Room Lights

0 W

0.5

20.5

Current

kWh today

kWh Month

OFF

Electric Fan

0 W

0.5

20.5

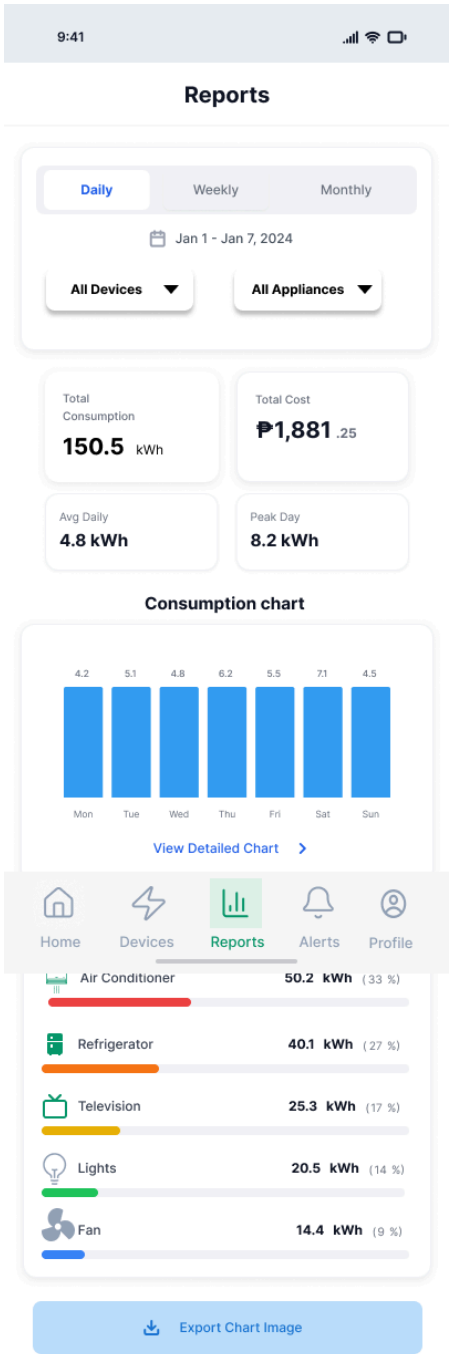
Current

kWh today

kWh Month

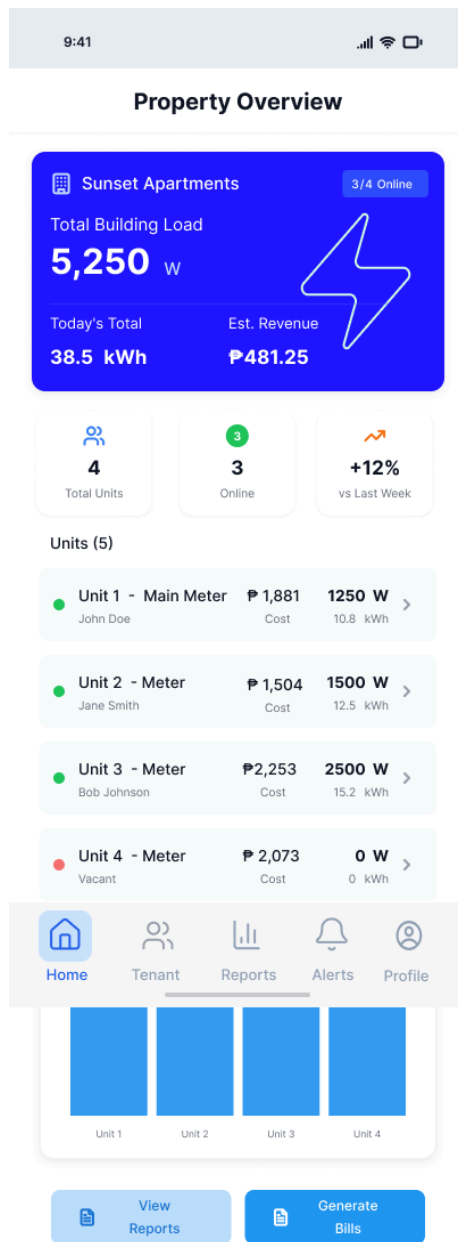
OFF

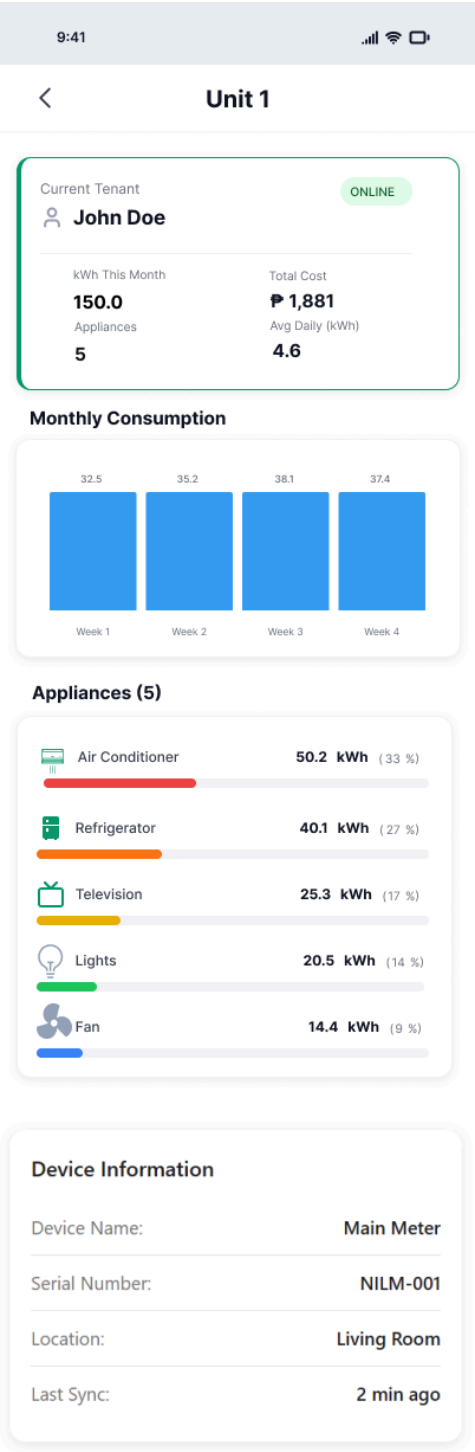
+





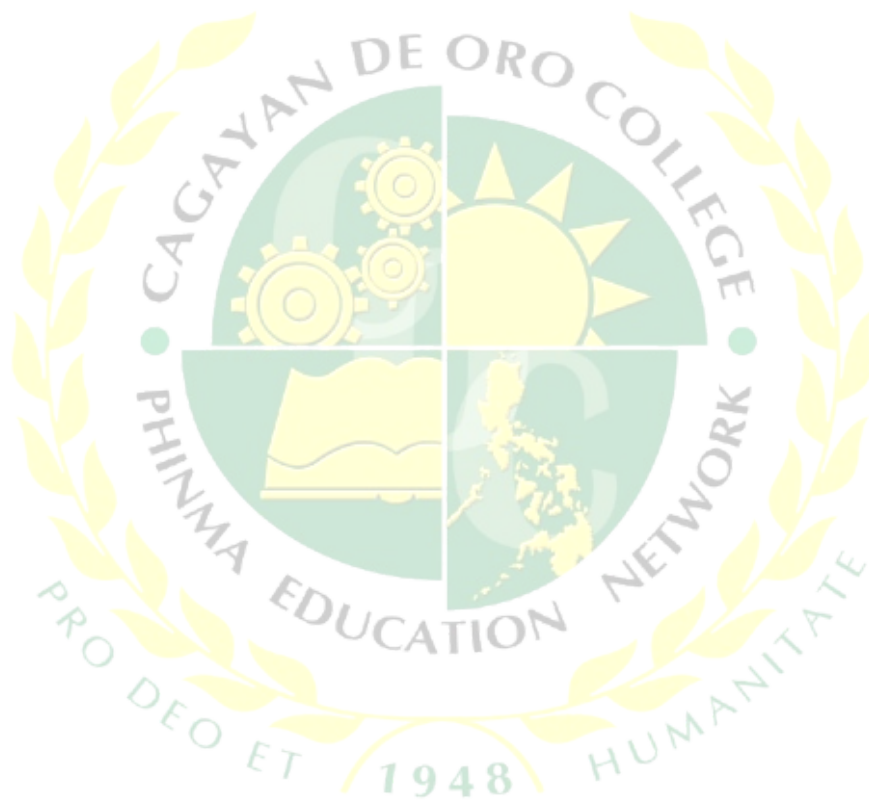
Dashboard







Appendix N: Detailed SUS Test



Appendix O: Functional and Non-Functional Requirements



CHAPTER 4



PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents the results obtained from the testing, validation, and evaluation of the developed Software-Based Non-Intrusive Load Monitoring (NILM) System for Residential Energy Monitoring. The findings include the verification of functional and non-functional requirements, software quality evaluation using the ISO/IEC 25010 software quality model, System Usability Scale (SUS) assessment, appliance classification accuracy testing, and system performance evaluation. The results are presented, analyzed, and interpreted to determine whether the developed system achieved the objectives of the study.

4.1 System Requirements Analysis

The evaluation of the developed Software-Based Non-Intrusive Load Monitoring (NILM) System began with a comprehensive verification of its functional and non-functional requirements. The analysis focused on validating the software components developed in this study, including the mobile application, backend services, database integration, appliance classification module, and visualization features.

The electrical monitoring hardware used for acquiring voltage, current, and power measurements was developed in collaboration with the Electrical Engineering team and served as the primary source of electrical data. However, the evaluation presented in this chapter primarily examines the software engineering implementation, specifically the system's ability to process incoming data, classify appliances, synchronize information with the database, generate reports, and provide real-time monitoring for tenants and landlords.

Through functional verification, integration testing, and operational validation, the developed software was assessed to determine whether it satisfied the system requirements identified during the requirements analysis phase.

4.1.1 Functional Requirements Verification Matrix

Functional requirements define the software operations that the developed Software-Based Non-Intrusive Load Monitoring (NILM) System must perform to satisfy user requirements. These software functions include user authentication, role-based access control, real-time energy monitoring, appliance classification, billing computation, report generation, alert notification, database synchronization, and remote outlet control through communication with the external monitoring device.

The external electrical monitoring hardware, developed in collaboration with the Electrical Engineering team, supplied the electrical measurements required by the application. However, the verification presented in this section focuses on validating the software modules responsible for processing, storing, analyzing, and presenting these data to end users.

To ensure complete traceability between the identified requirements and the implemented software modules, a Functional Requirements Verification Matrix was prepared. Each functional requirement was verified through functional testing, integration testing, and operational validation to determine whether the developed software satisfied its intended functionality.

Table 4.1 Functional Requirements Verification Matrix



Module ID	Target Functional Requirement	Technical Verification Status	Operational Outcome
FR-01	User Authentication		Users can securely login according to assigned roles.
FR-02	Tenant Dashboard		Displays real-time energy consumption and appliance usage.
FR-03	Landlord Dashboard		Displays unit-level monitoring and billing summaries.
FR-04	Real-Time Monitoring		Receives electrical readings from monitoring device.
FR-05	Appliance Classification		Successfully identifies appliance load signatures.
FR-06	Billing Computation		Automatically computes estimated electricity cost.
FR-07	Report Generation		Generates historical energy consumption reports.
FR-08	Remote Outlet Control		Sends ON/OFF commands to the monitoring device.

4.1.2 Non-Functional Requirements Verification

4.2 System Features and Functionalities

The completed Software-Based Non-Intrusive Load Monitoring (NILM) System provides an integrated software platform for residential energy monitoring. The system combines a mobile application, backend services, and a centralized database to enable real-time energy monitoring, appliance classification, historical data visualization, and billing management. Electrical measurements are obtained from the monitoring hardware developed in collaboration with the Electrical Engineering team, while the software developed in this study processes, stores, and presents the information to end users. The system features are divided into two primary user interfaces:

4.2.1 Tenant Mobile Application

- Login
- Dashboard
- Appliance Monitoring
- Reports
- Notifications
- Energy Cost

(SS sa system)

4.2.2 Landlord Monitoring Dashboard

- Unit Monitoring
- Billing
- Tenant Consumption
- Reports
- Appliance Summary

(SS sa system)

4.3 System Architecture and Design

This section presents the implemented software architecture and logical design of the developed Software-Based Non-Intrusive Load Monitoring (NILM) System. The architecture illustrates how the software components including the mobile application, backend server, database, and appliance classification module work together to process electrical measurements obtained from the monitoring hardware developed in collaboration with the Electrical Engineering team. While the hardware serves as the source of electrical data, the discussion in this section focuses on the software architecture responsible for data processing, storage, visualization, and user interaction.

4.3.1 Use Case Diagram (Behavioral Model)

Figure 4.1 Use Case Diagram Modeling Behavioral Actors, System Boundaries, and Core Functional Triggers

4.3.2 Process Model: Data Flow Diagram (DFD)

4.3.3 Data Model: Relational Schema Design

Figure 4.4. Relational Database Schema Design Representing Entity Attributes, DataTypes, and Third Normal Form (3NF) Key Constraints

4.4 System Evaluation

4.4.1 ISO/IEC 25010 Software Quality Assessment

The developed Software-Based Non-Intrusive Load Monitoring (NILM) System for Residential Energy Monitoring was evaluated using the ISO/IEC 25010 Software Quality Model. The assessment covered the eight software quality characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. Based on the evaluation results, the system obtained an overall rating of **[Overall Weighted Mean]**, interpreted as **[Interpretation]**. The findings indicate that the system meets the required software quality standards and is capable of providing reliable, efficient, secure, and user-friendly residential energy monitoring.

4.4.2 System Usability Scale (SUS) Results

The usability of the developed NILM system was evaluated using the System Usability Scale (SUS), a standardized tool for measuring user satisfaction and ease of use. The system obtained a SUS score of **[SUS Score]**, corresponding to an **[Interpretation]** usability rating. The results indicate that respondents found the system easy to learn, simple to use, and effective in monitoring residential energy consumption, demonstrating that it is suitable for practical implementation.

4.4.3 Pre- vs. Post-Automation Operational Metrics

4.5 Discussion of Results

CHAPTER 5:

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary of the findings, conclusions, and recommendations derived from the evaluation of the developed Software-Based Non-Intrusive Load Monitoring (NILM) System for Residential Energy Monitoring. The findings are based on the verification of the system requirements, software quality evaluation, usability assessment, and system performance analysis presented in Chapter 4. The chapter also highlights how the developed software contributes to improving residential energy monitoring through real-time visualization, appliance classification, and role-based energy management. The electrical monitoring hardware developed in collaboration with the Electrical Engineering team served as the source of electrical measurements; however, the conclusions and recommendations primarily focus on the software developed in this study.

5.1 Summary of Findings

The capstone project was conducted to design, develop, and evaluate a Software-Based Non-Intrusive Load Monitoring (NILM) System for Residential Energy Monitoring that provides real-time energy monitoring, appliance classification, and electricity consumption analysis for tenants and landlords. The developed software integrates a mobile

application, backend services, and a centralized database to process and visualize electrical measurements obtained from the monitoring hardware. The primary focus of this study was the design, development, and evaluation of the software components responsible for data processing, appliance classification, visualization, reporting, and user interaction.

Based on the functional verification, software quality evaluation, usability assessment, and system performance testing presented in Chapter 4, the major findings of the study are summarized as follows:

5.1.1 System Requirements and Functional Validation

- **Functional Alignment:** All identified functional requirements of the Software-Based Non-Intrusive Load Monitoring (NILM) System were successfully verified through functional testing, integration testing, and operational validation. The developed software correctly performed user authentication, real-time energy monitoring, appliance classification, billing computation, report generation, and notification services.
- **Operational Impact:** The developed software successfully processed electrical measurements received from the monitoring hardware and transformed them into meaningful energy consumption information for tenants and landlords through real-time dashboards, historical reports, and appliance **monitoring**.

5.1.2 Software Quality and Usability Metrics

- **ISO/IEC 25010 Evaluation:** The developed software achieved an overall mean score of ____, corresponding to a qualitative interpretation of Strongly Agree (Excellent). The evaluation indicates that the system satisfies the software quality characteristics defined in the ISO/IEC 25010 standard.



- System Usability Scale (SUS): The computed SUS score of ____ indicates that users found the developed software easy to learn, efficient to use, and suitable for residential energy monitoring.

5.1.3 Software Performance and Operational Improvements

- The developed software demonstrated reliable communication between the mobile application, backend server, and database.
- The appliance classification module successfully processed electrical measurements and identified supported household appliances.
- The system generated historical reports and billing summaries with minimal response delay.

5.2 Conclusions

Based on the empirical findings generated during the system evaluation, the following conclusions are drawn:

1. Achievement of the Study Objectives

The developed **Software-Based Non-Intrusive Load Monitoring (NILM) System** successfully achieved the objectives of the study by providing a software platform capable of processing electrical measurements, performing appliance classification, and delivering real-time residential energy monitoring through a mobile application. The system effectively supports tenants and landlords by providing accessible energy consumption information, historical reports, estimated electricity costs, and role-based monitoring features.

2. Software Architecture and System Performance

The integration of the React Native mobile application, Node.js backend services, and MySQL database resulted in a stable, efficient, and scalable software architecture capable of supporting real-time data processing, centralized data management, appliance classification, billing computation, and historical reporting. The implemented software architecture demonstrated reliable communication among the

mobile application, backend services, and database, ensuring consistent system performance throughout the evaluation period.

3. Software Quality and Collaboration

The results of the ISO/IEC 25010 software quality evaluation and the System Usability Scale (SUS) indicate that the developed software is functional, reliable, efficient, secure, and user-friendly for its intended users. Furthermore, the collaboration with the Electrical Engineering team enabled the integration of electrical monitoring hardware as the source of electrical measurements, while the Information Technology team successfully developed the software components responsible for data processing, appliance classification, backend integration, database management, visualization, and user interaction. This collaboration demonstrates the effective integration of hardware and software technologies in providing a comprehensive residential energy monitoring solution.

5.3 Recommendations

To ensure the long-term sustainability, scalability, and continuous improvement of the developed Software-Based Non-Intrusive Load Monitoring (NILM) System for Residential Energy Monitoring, the following recommendations are presented to stakeholders, system developers, and future researchers.

5.3.1 For Tenants and Landlords

- Utilize the developed system regularly to monitor energy consumption and identify appliances with high electricity usage.
- Use the generated reports and billing summaries to promote responsible energy consumption and improve transparency in electricity cost allocation.

5.3.2 For System Developers

- Improve the appliance classification algorithm by incorporating additional appliance load signatures and machine learning techniques to increase detection accuracy.
- Enhance the mobile application by adding advanced visualization tools, customizable notifications, and offline synchronization capabilities.

- Strengthen security through continuous updates, regular vulnerability assessments, and enhanced authentication mechanisms.

5.3.3 For Future Researchers

- Investigate the integration of advanced machine learning and deep learning techniques to improve appliance classification accuracy.
- Expand the developed software to support additional residential appliances and larger-scale residential deployments.
- Explore cloud-based analytics and predictive energy consumption models for smarter residential energy management.

